Efficient GPU-Accelerated Adaptive Minimum Cost Seed Selection [Technical Report]

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ABSTRACT

Efficient influence estimation and seed selection are crucial to social network advertising and are widely studied in data management. We focus on adaptive minimum cost seed selection (AMCSS), which selects *seed nodes* adaptively over multiple rounds, to reach a target number η of influenced users while minimizing total seed cost. Prior work, notably ASTI with multi-root reverse reachable sets (mRR-sets), remains computationally expensive on large graphs, often taking hours to days even with CPU parallelism.

We present GAAS, a GPU-accelerated method that integrates novel algorithmic and GPU-aware system designs to solve AMCSS in minutes. Processing mRR-sets involves irregular access patterns and variable-size samples, mismatching the GPU parallel architecture. Hence, we first develop a *GPU-tailored mRR-set structure*, GmRR, that assigns each thread block exclusive ownership of an equal-size *segment* with a *circular* layout, enabling efficient parallel mRR-set management while minimizing write contention. With GmRR, we design a GPU kernel ParallelGen to generate mRR-sets. Unlike prior work that regenerates mRR-sets from scratch in each round, we propose to update and reuse those from previous round, improving efficiency while requiring GPU-aware designs and rigorous theoretical analysis. Specifically, we design a ParallelUpdate kernel with *theoretically grounded update rules* that uses *circular segment updates* on GmRR for efficient mRR-set updates, together with a load-balancing scheme. We further devise a Select kernel for parallel seed selection. Integrating these together, GAAS efficiently solves AMCSS on GPUs with guarantees. Extensive experiments on large real-world graphs under different diffusion models show that GAAS is over an order of magnitude faster (up to 68.9×) than parallel CPU and GPU baselines, while the seed cost is among the lowest.

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1 INTRODUCTION

Social networks are a dominant medium for information dissemination, where users share content and influence with each other at scale. As of October 2025, over 5.6 billion people—more than two-thirds of the world's population—use social networks [10], driving a social advertising market exceeding $136 billion [3].

Advertisers select a set of influential users as *seeds*, each with a *cost*, to initiate diffusion process to influence a large number of users in the social network, to achieve a large *influence spread*. Identifying promising seeds under cost and time constraints requires accurate influence estimation, which is computationally expensive. This has motivated extensive work on scalable influence estimation and seed selection in data management [16, 17, 22–24, 33, 34, 41, 42, 45, 47].

A typical formulation specifies a target number of influenced users η (*influence threshold*) and seeks a seed set of minimum cost whose influence spread reaches η, known as the minimum cost seed selection (MCSS) problem [12, 18, 20]. MCSS is non-adaptive: seeds are selected in a single batch without observing actual diffusion, so the realized spread can deviate substantially from η [12, 18].

Therefore, *adaptive* MCSS (AMCSS) has been proposed to select seeds iteratively over multiple rounds, where each round's choices depend on observed diffusion outcomes [13, 14, 39, 44]. AMCSS is NP-hard, motivating the development of efficient approximate algorithms with theoretical guarantees. ASTI [39] is state of the art that introduces multi-root reverse reachable sets (mRR-sets) to estimate user influence via stochastic breadth-first-search (BFS) from randomly selected root nodes. However, solving AMCSS on large graphs remains computationally prohibitive. ASTI often takes tens of thousands of seconds—more than one *day* for large η on large graphs—while a parallel CPU version of ASTI still needs many thousands of seconds, around half a day. Such latency severely limits the practicality of AMCSS in time-critical applications.

We aim to improve the processing time to *minutes* (hundreds of seconds) while preserving result quality. This is challenging and existing methods largely ignore system design aspects. We achieve this by proposing a GPU-Accelerated AMCSS method, GAAS, which integrates novel algorithmic designs with GPU-aware optimizations to exploit the massive parallelism of modern GPUs.

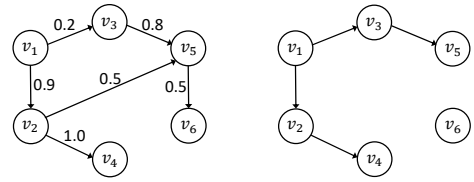
GPUs are increasingly important for general-purpose parallel computing [21, 36, 38, 48–50]. Compared with CPUs, GPUs have a fundamentally different architecture. They execute massive numbers of lightweight threads in a SIMT (Single Instruction Multiple Thread) model, organized into blocks and warps, which suits data-parallel tasks with simple control flow and regular memory access but performs poorly under complex branching and irregular access patterns. The memory hierarchy also differs significantly, and dynamic memory allocation within kernels incurs high overhead. Therefore, directly porting CPU-native algorithms to GPUs often results in suboptimal performance.

Specifically, the main cost in AMCSS is sampling and processing mRR-sets in each adaptive round for influence estimation. This involves irregular graph traversals and variable-size samples: the size of each mRR-set is unknown a priori and the required number of mRR-sets also changes by round, complicating GPU data structures and parallelization. Moreover, existing methods regenerate all mRR-sets from scratch in every round, wasting computation despite the close relationship of mRR-sets between successive rounds.

In contrast, GAAS updates and reuses mRR-sets from the previous round, which is more efficient but requires careful GPU-aware designs and theoretical analysis to ensure the updated mRR-sets are equivalent to newly generated ones for accurate influence estimation. To this end, we first design a *GPU-tailored mRR-set structure*, GmRR, that supports efficient storage, parallel generation, and update/reuse of mRR-sets across rounds in GAAS. Specifically, GmRR partitions a pre-allocated global array into disjoint equal-size *segments*, each exclusively owned by one GPU thread block, minimizing write contention in parallel. Within each segment, a *circular* layout allows variable-length mRR-sets to be appended and updated with little data movement. All components in GmRR are flat arrays with regular access patterns, enabling efficient GPU execution. With the GmRR structure, we design GPU kernels to parallelize mRR-set generation, update, and seed selection in GAAS. The `ParallelGen` kernel generates mRR-sets by evenly partitioning them across thread blocks and performing warp-centric parallel BFS within each block. We then develop the `ParallelUpdate` kernel to update mRR-sets from the previous round. We first derive *theoretically sound update rules* for reusing mRR-sets under changes in the graph and influence threshold, and then parallelize the update by letting each block update the mRR-sets in its own segment using a *circular segment update* on GmRR that enables in-segment updates and appends without excessive data movement, together with a load-balancing technique. With the mRR-sets ready, we devise a GPU kernel `Select` that selects each seed batch in parallel and updates all related data structures for the next round. Integrating these components, GAAS solves AMCSS with an approximation ratio of $\ln \eta / [(1 - 1/e)(1 - \epsilon)]$, improving over existing study, and achieves high GPU efficiency.

Extensive experiments on real-world data under different diffusion models show that GAAS is typically over an order of magnitude faster than the baselines, while yielding seeds with comparable or lower total cost. Our contributions are summarized as follows.

- We design a GPU-accelerated method GAAS for solving AMCSS that significantly improves efficiency by integrating novel algorithmic designs with GPU-specific optimizations.



(a) Social network G (b) Realization ϕ

Figure 1: A social graph and a possible realization.

- We develop a GPU-tailored mRR-set structure GmRR that supports efficient storage, generation, and update of mRR-sets on GPUs, while reducing write contention and improving parallelism.
- We propose theoretically grounded mRR-set update rules across rounds, and GPU kernels with circular segment update on GmRR for efficient mRR-set update, generation, and seed selection.
- Extensive experiments show that GAAS is often over an order of magnitude faster than baselines while maintaining low seed cost under different diffusion models.

2 PRELIMINARIES

2.1 Influence Diffusion Models

The social network is modeled as a directed graph $G = (V, E)$ with $|V| = n$ users and $|E| = m$ directed edges. Each edge $e = (u, v) \in E$ represents a social tie from u to v along which influence may diffuse. The in- and out-neighbor sets of u are denoted by $N_{\text{in}}(u)$ and $N_{\text{out}}(u)$, respectively. Each edge e has a *diffusion probability* $p(e) = p(u, v) \in [0, 1]$, and each node $u \in V$ has a positive *seed cost* $c(u)$, which we normalize to $(0, 1]$ by dividing all costs by the maximum. Figure 1(a) shows a graph G, where the number on each edge indicates its diffusion probability.

On graph G, given a *seed set* $S \subseteq V$, influence diffusion follows the widely adopted models, *independent cascade* (IC) or *linear threshold* (LT) [4, 39–41], in discrete steps. At step 0, only nodes in S are active, while all other nodes are inactive. When a node u becomes active at step t, it has one chance in step $t+1$ to activate each inactive out-neighbor; afterwards, u makes no further attempts. The two models differ in how u activates an out-neighbor v. In the IC model, u activates v independently with probability $p(u, v)$. In the LT model, each node v has a threshold $\lambda_v \sim U(0, 1)$, and $p(u, v)$ is treated as a weight; v becomes active if the total weight from its active in-neighbors is no less than λ_v, i.e., $\sum_{u \in N_{\text{in}}(v)} \mathbb{I}_{\{u \text{ is active}\}} p(u, v) \geq \lambda_v$, where $\mathbb{I}_{\{\cdot\}}$ is the indicator function. The diffusion process stops when no new node is activated, and the total number of activated nodes is denoted by $I(S)$, which is the *influence spread* of S.

Note that $I(S)$ is a random variable due to the randomness in the diffusion process, which may lead to many possible outcomes. The IC propagation process can be equivalently described by the live-edge procedure [25]. Specifically, for each edge $e = (u, v) \in E$, we independently flip a coin with head probability $p(e)$ to decide whether e is *live* or *blocked*. All blocked edges are removed and the remaining graph is referred to as a *realization* of the probabilistic graph G, denoted by ϕ. Under a realization ϕ, the influence spread of a seed set S, denoted as $I_\phi(S)$, is the number of nodes reachable from S through live edges. For example, Figure 1(b) shows a realization ϕ with live edges marked while blocked edges removed in the IC model. Diffusing from seed v_2 over ϕ yields $I(\{v_2\}) = 2$, since v_2

can reach v_4 and itself. Similarly, for the LT model, a realization ϕ is obtained by sampling a threshold λ_v for each node v. The influence spread of S under ϕ is then the number of nodes that become active from S according to the sampled thresholds.

Let Ω be the set of all possible realizations and $p(\phi)$ denote the probability of a specific realization ϕ. For ϕ, the influence spread $I_\phi(S)$ is deterministic. Thus, the expected spread is $\mathbb{E}_\Phi[I(S)] = \sum_{\phi\in\Omega} I_\phi(S) \cdot p_\phi$, which is the weighted average spread over all possible realizations in Ω.

2.2 Adaptive Minimum Cost Seed Selection

Conventional non-adaptive MCSS identifies a seed set S with minimum cost such that $\mathbb{E}_\Phi[I(S)] \geq \eta$, without knowing the actual realization in advance [12, 18, 51]. As a result, the realized number of activated users may be far fewer than η in some realizations, or significantly more than η in others.

In contrast, the adaptive setting AMCSS selects seeds over multiple rounds, with each round's choices based on the observed diffusion outcomes so far. This feedback enables more cost-effective decisions and keeps the realized spread close to the target η, making AMCSS more practical in real deployments. Specifically, in round i, we select a batch of b seeds S_i on G_i, trigger diffusion from S_i, and observe the activated nodes A_i (including S_i). We then update the remaining influence threshold to $\eta_{i+1} = \eta_i - |A_i|$ and remove A_i and their incident edges to obtain the residual graph $G_{i+1} = (V_{i+1}, E_{i+1})$. Initially, $G_1 = G$, $V_1 = V$, $E_1 = E$, and $\eta_1 = \eta$. The process stops once at least η users are activated, and the final seed set is $S = \bigcup_i S_i$.

An adaptive policy π specifies how seeds are selected in each round. For a realization ϕ, let $S(\pi, \phi)$ be the seeds chosen by π, and let $C(\pi, \phi) = \sum_{v\in S(\pi,\phi)} c(v)$ be their total cost. The AMCSS problem is formally given in Definition 2.2; it is NP-hard by reduction from adaptive seed minimization (ASM) with unit costs $c(v) = 1$ for all v, which is NP-hard [39].

Definition 2.1 (The AMCSS Problem). Given a graph $G = (V, E)$ with propagation probability $p(e)$ for each $e \in E$, cost $c(v)$ for each $v \in V$, and an influence threshold η, AMCSS seeks a policy π^* that minimizes the expected cost of selected seeds while ensuring the influence spread is at least η on any realization $\phi \in \Omega$. That is,

$$\pi^* = \text{argmin}_\pi \ \mathbb{E}_\Phi[C(\pi, \Phi)] \quad \text{s.t. } |I(S(\pi, \phi))| \geq \eta, \forall \phi \in \Omega.$$

Since AMCSS only requires reaching η, we use the *truncated influence spread* $\Gamma_\phi(S) = \min\{I_\phi(S), \eta\}$ for each realization ϕ [39]. In round i, given previous seeds $\mathcal{S}_{i-1} = \bigcup_{j=1}^{i-1} S_j$ that activated $I_\phi(\mathcal{S}_{i-1})$ nodes, the *marginal truncated influence* of $S_i \subseteq V_i$ is defined as $\Gamma_\phi(S_i|\mathcal{S}_{i-1}) = \Gamma_\phi(S_i \cup \mathcal{S}_{i-1}) - \Gamma_\phi(\mathcal{S}_{i-1})$. Over all realizations, the expected versions are $\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})] = \sum_\phi p_\phi \Gamma_\phi(S|\mathcal{S}_{i-1})$ and $\mathbb{E}[\Gamma(S)] = \sum_\phi p_\phi \Gamma_\phi(S)$.

2.3 Graphics Processing Units

GPUs are widely used for general-purpose parallel computing [5, 28, 29, 36, 38]. The most popular GPU programming framework is Nvidia CUDA, which provides a C/C++-like programming interface for developers to harness the computational power of Nvidia GPUs.

When a GPU *kernel* is launched, the GPU creates a grid of threads organized in a hierarchical structure: the *grid* partitions threads into thread *blocks* of equal size (e.g., 1024 threads per block), and each block further divides into *warps*, typically containing 32 threads each. We employ one-dimensional grid with grid size and block size denoted by *gridDim.x* and *blockDim.x*, respectively. Grid size is the number of blocks, and block size is the number of threads in the block. GPUs follow the Single Instruction Multiple Thread (SIMT) execution model, where all threads within a warp execute the same instruction in lockstep. This makes GPUs particularly efficient for data-parallel tasks with simple control flow, in contrast to CPU threads which can handle complex branching logic independently.

GPUs also feature a hierarchical memory architecture. The *global memory* is the largest memory accessible by all threads, while shared memory is a smaller memory shared among threads within the same block. Dynamic memory allocation within kernels is supported through the device heap. However, device heap allocations incur substantial overhead and cause fragmented memory layouts. Since multiple threads may update the same memory location concurrently, GPU programs must use atomic operations judiciously. Excessive usage of such primitives may hinder efficiency.

Therefore, GPUs excel at processing simple data structures (e.g., arrays) with regular access patterns (e.g., sequential or strided), but perform poorly on complex data structures and irregular random accesses. Consequently, straightforward parallelization of CPU algorithms on GPUs often yields suboptimal performance due to this mismatch. GPU-accelerated algorithms must therefore be carefully redesigned to leverage simple data layouts and regular memory access patterns for efficient execution.

Algorithm 1: ASTI

Input: Graph G, target η, estimation error ϵ, batch size b, cost $c(\cdot)$.
Output: Seed set S.

$i \leftarrow 1, G_i \leftarrow G, \eta_i \leftarrow \eta, \rho_b \leftarrow 1 - (1 - 1/b)^b, S \leftarrow \emptyset$;
while $\eta_i > 0$ **do**
  $\delta \leftarrow 1/n_i$; $\hat{\epsilon} \leftarrow (\epsilon - \delta)(1 - \delta)$; $S_i \leftarrow \emptyset$;
  Estimate θ and $\theta_{\max}$; $\theta_i \leftarrow \theta$; $H \leftarrow \lceil \log_2(\theta_{\max}/\theta) \rceil + 1$;
  $a_1 \leftarrow \ln(3H/\delta) + \ln \binom{n_i}{b}$; $a_2 \leftarrow \ln(3H/\delta)$;
  while $\theta_i < \theta_{\max}$ **do**
    Generate additional mRR-sets until total reaches θ_i;
    while $|S_i| < b$ **do**
      $u \leftarrow \text{argmax}_{v\in V_i} \frac{\Lambda(S_i \cup \{v\}) - \Lambda(S_i)}{c(v)}$;
      $S_i \leftarrow S_i \cup \{u\}$;
    $\Lambda^l(S_i) \leftarrow \left(\sqrt{\Lambda(S_i) + 2a_1/9} - \sqrt{a_1/2}\right)^2 - a_1/18$;
    $\Lambda^u(S_i^o) \leftarrow \left(\sqrt{\Lambda(S_i)/\rho_b + a_2/2} + \sqrt{a_2/2}\right)^2$;
    if $\Lambda^l(S_i) > \rho_b(1 - \hat{\epsilon})\Lambda^u(S_i^o)$ **then break**;
    $\theta_i \leftarrow 2\theta_i$;
  $S \leftarrow S \cup S_i$;
  Diffuse from S_i, and obtain the activated nodes A_i;
  $G_{i+1} \leftarrow G_i \backslash A_i$; $\eta_{i+1} \leftarrow \eta_i - |A_i|$; $i \leftarrow i + 1$;

2.4 mRR-set and Existing Solutions

Here we review the most relevant work, leaving other related work to Section 4. The work of [13, 14] is the first to study the AMCSS problem, proposing a greedy policy that, in each round, selects the seed with the maximum influence-to-cost ratio and achieves an approximation ratio of $(\ln \eta + 1)^2$. However, this policy assumes

exact knowledge of truncated influence, which is #P-hard to compute [7]. To address this, Tang et al. introduce the *multi-root Reverse Reachable* set (mRR-set) to efficiently estimate truncated influence and design the ASTI method [39] outlined below.

In each round i, an mRR-set on the residual graph G_i is generated as follows: (i) sample $k_i = n_i/\eta_i$ roots uniformly at random from V_i; (ii) run a stochastic reverse BFS from these roots, where each edge (u, v) is traversed with probability $p(u, v)$; and (iii) take all visited nodes (including the roots) as the mRR-set R. Since k_i may be non-integer, we realize it in expectation by sampling $\lceil k_i \rceil$ roots with probability $k_i - \lfloor k_i \rfloor$ and $\lfloor k_i \rfloor$ roots otherwise. A node set S *covers* an mRR-set R if $S \cap R \neq \emptyset$, and its *coverage* $\Lambda(S)$ is the number of mRR-sets it covers. Given θ mRR-sets, we estimate $\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})]$ by $\hat{\Gamma}(S|\mathcal{S}_{i-1}) = \eta_i \Lambda(S)/\theta$, which is shown in [39] to approximate $\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})]$ with bounded error:

$$(1 - 1/e)\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})] \leq \mathbb{E}[\hat{\Gamma}(S|\mathcal{S}_{i-1})] \leq \mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})],$$

where expectation $\mathbb{E}[\hat{\Gamma}(S|\mathcal{S}_{i-1})]$ is over mRR-set randomness.

The original ASTI assumes uniform seed cost, i.e., $c(v) = 1$ for all $v \in V$ [39]. Algorithm 1 extends ASTI to non-uniform costs. It iteratively selects seeds and triggers diffusion until the influence threshold η is reached (Line 3). In each round i, ASTI selects a batch of b seeds S_i that approximates the optimal batch S_i^o within a factor $\rho_b(1-1/e)(1-\epsilon)$, where $\rho_b = (1-1/b)^b$ and ϵ are the estimation error parameters initialized at Line 1. Initially, ASTI generates a small number θ_i of mRR-sets. Using these sets, it greedily selects b seeds with the largest coverage-to-cost ratio (Lines 8–10), removing the mRR-sets they cover after each selection. It then computes a lower bound $\Lambda^l(S_i)$ on the coverage of S_i and an upper bound $\Lambda^u(S_i^o)$ on the optimal coverage (Lines 11–12). If $\Lambda^l(S_i)$ approximates $\Lambda^u(S_i^o)$ within the required ratio, S_i is accepted, added to S (Line 15), and influence diffusion is triggered (Line 16), updating the residual graph and influence target for the next round at Line 17. Otherwise, θ_i is doubled (Line 14), and the process from Lines 7–14 is repeated; if θ_i reaches the maximum $\theta_{\max}$, S_i is also accepted. Overall, ASTI achieves an approximation ratio of $(\ln \eta + 1)^2/[(1 - 1/e)(1 - \epsilon)]$.

ASTI is a milestone for solving the AMCSS problem, but in each adaptive round it regenerates mRR-sets from scratch, incurring substantial overhead. Even with CPU parallelization, ASTI still needs thousands of seconds on real graphs. Our goal is to reduce this to hundreds or even tens of seconds by exploiting GPUs. As discussed in Section 2.3, GPUs have fundamentally different compute and memory architectures, so achieving this requires new algorithmic designs and GPU-specific optimizations tailored to AMCSS.

3 GPU-ACCELERATED METHOD GAAS

3.1 Overview

GAAS is a GPU-centric framework for AMCSS, tightly coupling novel algorithmic designs with GPU-friendly mRR-set processing.

The dominant computational cost in solving AMCSS lies in sampling mRR-sets for influence estimation. On GPUs, efficiently managing these irregular, dynamically evolving samples is challenging because dynamic memory allocation is inefficient and global synchronization is expensive, making direct ports of CPU-based designs ineffective. Furthermore, to avoid regenerating mRR-sets from scratch in every adaptive round, we aim to reuse and update them across rounds, which introduces additional challenges in theoretical analysis, algorithm design, and GPU-parallel optimization.

To this end, we propose GmRR, a GPU-tailored structure for storing, managing, and updating mRR-sets (Section 3.2). GmRR reconciles the irregular, evolving nature of mRR-sets with massive GPU parallelism via a specialized global memory layout. Specifically, GmRR utilizes a pre-allocated global array partitioned into disjoint *segments* of equal size, each mapped exclusively to a specific thread block. This ownership enables blocks to generate and update mRR-sets locally, eliminating expensive dynamic memory allocation and excessive atomic contention. Internally, each segment functions as a *circular* buffer, allowing variable-length mRR-sets to be appended and updated efficiently within a segment, supporting the update of mRR-sets across rounds without frequent costly data shifting.

To efficiently generate mRR-sets on GPU, with GmRR, we propose the ParallelGen kernel (Section 3.3). At the inter-block level, it evenly partitions mRR-set generation across thread blocks, which write mRR-sets into their own segments; at the intra-block level, it employs a GPU kernel Probe to perform stochastic BFS, where threads cooperatively traverse the nodes to construct an mRR-set.

To enable reuse of mRR-sets across rounds, we develop the GPU kernel ParallelUpdate with its supporting theory for mRR-set update (Section 3.4). An mRR-set from round $i-1$ reused in round i is *polluted* if it contains any node activated in round $i-1$, and *clean* otherwise. Both types must be carefully updated to preserve the theoretical properties of mRR-sets for accurate influence estimation: clean mRR-sets are augmented with additional roots to satisfy the sampling requirements on the residual graph G_i, whereas polluted mRR-sets are repaired by removing activated nodes and resampling roots together with their visited nodes. Building on these theoretical insights, we design the ParallelUpdate kernel over GmRR to implement mRR-set updates on GPUs efficiently. A key technique is *circular segment update* that treats each segment as a circular buffer, enabling efficient updates and appends within the segment, and eliminating expensive cumulative element shifts on the GPU. Moreover, we design a load-balancing scheme to mitigate inter-block workload skew from uneven numbers of polluted mRR-sets, further improving the efficiency of ParallelUpdate.

Besides, we devise a GPU Select kernel (Section 3.5) for seed selection. It uses a parallel reduction to find the node with the maximum coverage-to-cost ratio, and marks all mRR-sets containing it as covered while decrementing the coverage of nodes in them. After selecting b seeds, Select evaluates confidence bounds to decide whether the batch is accepted or more mRR-sets are required.

Section 3.6 integrates these to the full GAAS method, which orchestrates mRR-set generation, updates, and seed selection across adaptive rounds using GPU-efficient designs.

3.2 GPU-Tailored mRR-set Structure GmRR

A naive idea to parallelize the algorithms is assigning each GPU thread the task of generating a full mRR-set, including root sampling and stochastic BFS. However, this results in highly irregular and heavy workloads per thread, leading to poor GPU utilization.

Another approach is block-level parallelization, where all threads in a block construct one mRR-set. However, this still incurs substantial atomic and memory-management overhead: the mRR-set

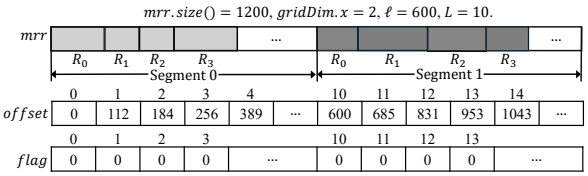


Figure 2: Example of GmRR with mrr, $offset$ and $flag$ arrays.

size is unknown until completion, multiple blocks may finish simultaneously and contend on a global write pointer. Accordingly, the mRR-sets need to be buffered via device-heap allocations before being copied to global memory, which is inefficient on GPUs.

Furthermore, we aim to update and reuse existing mRR-sets from the previous round $i-1$ when selecting seeds in the current round i, improving overall efficiency. Hence, we design GmRR, a GPU-tailored structure for storing mRR-sets. At its core, GmRR uses a *segment*-based layout where each thread block is assigned a dedicated, non-overlapping segment in global memory to store its generated mRR-sets. Each block stores its mRR-sets contiguously in its own segment, and segments are mutually disjoint. This removes both the global write pointer (each block writes only to its segment) and temporary buffer allocations, since all elements are written directly to global memory. In addition, GmRR is implemented as flat *arrays*, enabling efficient access on the GPU. The per-block segments further enable dynamic updates of mRR-sets via a *circular layout within each segment*, which we will describe in more detail in Section 3.4.

Figure 2 shows a GmRR instance with two segments, each owned by one thread block. GmRR consists of three arrays: mrr stores all mRR-sets, $offset$ records start positions of mRR-sets in mrr, and $flag$ marks whether an mRR-set is covered/polluted or not, to be used in subsequent sections. The total memory footprint of mRR-sets over all adaptive rounds is hard to estimate a priori, so we pre-allocate mrr as a large buffer constrained by the available GPU memory. Extending the design to settings where mRR-set storage exceeds a single GPU's memory, e.g., via graph partitioning and a multi-GPU GmRR, is left for future work.

Then we partition mrr into $gridDim.x$ non-overlapping segments, each of size ℓ, where $gridDim.x$ is the number of launched thread blocks. For example, the size of mrr is 1200 and ℓ is 600 in Figure 2. Each segment j stores the mRR-sets generated by block j, starting at position $j \cdot \ell$ and ending at $(j+1) \cdot \ell - 1$ in mrr. The segment size ℓ is set by dividing the total size of mrr by $gridDim.x$. As explained in Section 2.4, in round i each mRR-set has k_i roots and a variable number of nodes reached by reverse BFS. We store it as a contiguous sequence of node IDs: the first k_i entries are the roots, followed by the visited nodes. Each block uses all its threads to generate mRR-sets one by one and stores them contiguously within its assigned segment. For example, in Figure 2, Segment 0 (gray) stores the four mRR-sets R_0–R_3 generated by Block 0, and Segment 1 (dark) stores those generated by Block 1.

In GmRR, mRR-sets in a segment are always stored contiguously. Thus, each mRR-set can be identified by the start positions of two contiguous mRR-sets in the $offset$ array. For example, in Segment 0 the start positions of R_2 and R_3 are 184 and 256 at $offset[2]$ and $offset[3]$; the nodes in R_2 occupy from $mrr[184]$ to $mrr[255]$. The last mRR-set has its end position stored in $offset[r+1]$, where r is the number of mRR-sets in the segment. For example, the end position of R_3 in Segment 0 is at $offset[4]$, which is also the start position of free space in the segment for future mRR-sets.

Given the required number of mRR-sets per round, we evenly assign them to segments, simplifying in-segment lookup and balancing workload across segments and blocks. The remaining question is how many offset values to store in $offset$ for each segment. The number of mRR-sets per segment may vary across rounds, but we do not want to reconstruct $offset$ when this number changes. Since the global maximum number of mRR-sets over all rounds has an upper bound and the number of segments is fixed, we predefine a sufficiently large per-segment capacity L and reserve L offset entries per segment. The size of the $offset$ array is then $L \cdot gridDim.x$, since each segment corresponds one-to-one with a thread block. Every L entries in $offset$ record the start positions of at most L mRR-sets in the corresponding segment. The start position of the r-th mRR-set in the j-th segment is given by $offset[r + j \cdot L]$. For example, in Figure 2, L is set to 10, and thus the start position of the mRR-set R_1 in Segment 1 is $offset[1 + 1 \cdot L] = offset[11]$.

Accordingly, the $flag$ array also has $L \cdot gridDim.x$ entries; entry $(r + j \cdot L)$ marks whether the r-th mRR-set in segment j is covered/polluted or not. Section 3.5 details how $flag$ is updated.

3.3 Parallel mRR-set Generation

In round i, GAAS reuses and updates the θ_{i-1} mRR-sets from round i–1. (Section 3.4). If the required number θ_i exceeds θ_{i-1}, we generate $\theta_i - \theta_{i-1}$ additional mRR-sets. Especially, in round 1, we generate θ_1 mRR-sets from scratch, by applying the GPU kernel ParallelGen, where each block uses all its threads to construct new mRR-sets and write them into its own segment in GmRR.

The ParallelGen Kernel. Algorithm 2 presents the ParallelGen kernel for mRR-sets generation in a block $block.id$. In GPUs, block IDs are assigned contiguously from 0 to $gridDim.x - 1$, which we also use to index the segments in GmRR. As mentioned, when θ mRR-sets are needed, we let each segment store on average $\bar{\theta} = \theta / gridDim.x$ mRR-sets. Often, $\bar{\theta}$ is not an integer. In this case, we require the first $\theta \% gridDim.x$ segments to store $\lceil \bar{\theta} \rceil$ mRR-sets, and the remaining segments to store $\lfloor \bar{\theta} \rfloor$ mRR-sets. Therefore, in Line 1 we compute the number of existing mRR-sets r in the segment owned by block $block.id$ as $\theta_{i-1}/gridDim.x$, plus one if $block.id < \theta_{i-1} \% gridDim.x$. Likewise, in round i with θ_i mRR-sets required, we set the target number r' of mRR-sets in this segment to $\theta_i / gridDim.x + \mathbb{I}_{\{\theta_i \% gridDim.x > block.id\}}$. The difference $r' - r$ is the number of new mRR-sets that block $block.id$ generates in round i and stores in its segment. In Line 3, we set the starting write position b_w for the next mRR-set as the end of the last mRR-set from round $i-1$, *i.e.*, $offset[r + block.id * L]$. We also initialize $b_r \leftarrow b_w$, where b_r is the read pointer used by Probe (Algorithm 3) for stochastic BFS within the current mRR-set.

Each iteration of the while loop (Lines 4–12) constructs one mRR-set using all threads in the block. We first mark all nodes in the residual graph G_i as unvisited in parallel (Line 5). Next, we uniformly and parallelly sample k_i distinct roots from V_i (Line 7): thread j samples a root u and writes it to mrr at position $(b_w + j) \% \ell + block.id \cdot \ell$ (Line 8). This modulo-based address within the segment enables circular updates later. Line 9 atomically increments $cnt[u]$, the *coverage count* of node u (the number of mRR-sets containing u). We

Algorithm 2: The `ParallelGen` Kernel

Input: Graph G_i, coverage counter cnt, mrr, $offset$, θ_{i-1}, θ_i.

1 $r \leftarrow \theta_{i-1}/gridDim.x + \mathbb{I}_{\{(\theta_{i-1} \% gridDim.x) > block.id\}}$;
2 $r' \leftarrow \theta_i/gridDim.x + \mathbb{I}_{\{(\theta_i \% gridDim.x) > block.id\}}$;
3 $b_w \leftarrow offset[r + block.id * L]$; $b_r \leftarrow b_w$;
4 **while** $r < r'$ **do**
5 $\quad \forall u \in V_i$, $visited[u] \leftarrow false$ in **parallel**;
6 $\quad$ **parallel for** $j = thread.id$; $j < k_i$; j += $blockDim.x$ **do**
7 $\quad\quad$ Atomically sample $u \in V_i$ with $visited[u] == flase$ and set $visited[u] \leftarrow true$;
8 $\quad\quad mrr[(b_w + j) \% \ell + block.id * \ell] \leftarrow u$;
9 $\quad\quad$ atomicAdd($cnt[u]$, 1);
10 $\quad$ __syncthreads();
11 $\quad$ `Probe`(G_i, k_i, b_r, b_w, mrr, $visited$, cnt);
12 $\quad r \leftarrow r + 1$; $offset[r + block.id * L] \leftarrow b_w$; $b_r \leftarrow b_w$;

say that u covers an mRR-set if it appears in that set. After all k_i roots are written, `Probe` (Line 11) performs reverse BFS from these roots, appending newly visited nodes to the mRR-set and updating b_w, mrr, and cnt. Finally, we advance r, record the end position of the new mRR-set in $offset[r + block.id \cdot L] \leftarrow b_w$, and set $b_r \leftarrow b_w$ to start the next mRR-set generation (Line 12).

The `Probe` Kernel. Given arbitrary k roots, `Probe` performs reverse BFS from the roots to complete the mRR-set. b_r denotes the start of the current mRR-set and is used to index frontier nodes whose in-neighbors are probed. b_w denotes the next write position in mrr within the segment of $block.id$. We maintain a BFS queue indexed by $head$ and $tail$, initialized to 0 and k respectively, so that $tail - head$ equals the number of frontier nodes pending expansion. To better utilize threads, each warp cooperatively processes the in-neighbors of a single frontier node, while different warps handle different frontier nodes. The lane identifier $lane.id$ of each thread is $thread.id$ mod 32. The loop in Line 2 proceeds while the BFS queue is non-empty. In each iteration, the first thread in the warp ($lane.id = 0$) obtains the frontier index idx (Line 3), which is then broadcast to all threads in the warp (Line 4). If idx is valid, i.e., smaller than $tail$ (Line 5), each warp retrieves the frontier node u (Line 6). The threads in warp iterate over a subset of u's unvisited in-neighbors v (Lines 7–8), and check whether edge (v, u) satisfies the IC or LT activation condition in Section 2.1 (Line 9). If so, v is appended to the mRR-set at the next circular position in mrr by atomically incrementing b_w (Line 10), and we update $visited[v]$, $cnt[v]$, and the BFS queue tail (Line 11).

3.4 Parallel mRR-set Update

In round i, we reuse from round $i-1$. After selecting seeds S_{i-1} and simulating diffusion, we obtain activated nodes A_{i-1} and form the residual graph G_i by removing A_{i-1} and their incident edges from G_{i-1}. An mRR-set from round $i-1$ is *polluted* if it contains any node in A_{i-1}; otherwise it is *clean*. To reuse the mRR-sets in round i, we must update both clean and polluted ones so that each updated mRR-set has exactly the same distribution as an mRR-set generated directly on G_i, thereby preserving its theoretical guarantee. In round i, each mRR-set must contain $k_i = n_i/\eta_i = (n_{i-1} - |A_{i-1}|)/(\eta_{i-1} - |A_{i-1}|)$ roots, and k_i is monotonically non-decreasing over rounds (*i.e.*, $k_i \geq k_{i-1}$), since $|A_{i-1}| \geq 1$ and $n_i \geq \eta_i$.

Algorithm 3: The `Probe` Procedure

Input: Graph G_i, root number k, b_r, b_w, mrr, $visited$, cnt

1 $b_w \leftarrow b_r + k$; $head \leftarrow 0$; $tail \leftarrow k$; $lane.id \leftarrow thread.id \% 32$;
2 **while** $head < tail$ **do**
3 $\quad$ **if** $lane.id == 0$ **then** $idx \leftarrow$ atomicAdd($head$, 1);
$\quad\quad$ __syncwarp();
4 $\quad$ Broadcast idx to threads in the same warp;
5 $\quad$ **if** $idx < tail$ **then**
6 $\quad\quad u \leftarrow mrr[(b_r + idx) \% \ell + block.id * \ell]$;
7 $\quad\quad$ **parallel for** $j = lane.id$; $j < |N_{in}(u)|$; j += 32 **do**
8 $\quad\quad\quad$ Get the j-th node $v \in N_{\text{in}}(u)$ with $visited[v] == flase$ and set $visited[v] \leftarrow true$;
9 $\quad\quad\quad$ **if** $p(v, u)$ *meets the condition in IC or LT* **then**
10 $\quad\quad\quad\quad mrr[$atomicAdd$(b_w, 1) \% \ell + block.id * \ell] \leftarrow v$;
11 $\quad\quad\quad\quad$ atomicAdd($cnt[v]$, 1); atomicAdd($tail$, 1);
12 $\quad\quad$ __syncwarp();

Hence even clean mRR-sets must be updated to satisfy the new root requirement. Moreover, updating mRR-sets in `GmRR` requires dedicated designs to avoid excessive data movement on the GPU.

Below, we first analyze updates of clean and polluted mRR-sets, then present a *circular segment update* strategy over `GmRR`, and finally introduce the `ParallelUpdate` kernel, which performs these updates in parallel while preserving the `GmRR` layout.

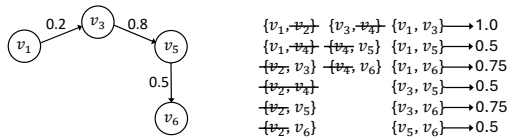
Given the activated node set A_{i-1} from round $i-1$, we first identify polluted mRR-sets in `GmRR` by invoking the `Discover` kernel (Algorithm 6) with A_{i-1} as input: every mRR-set containing a node in A_{i-1} is marked as polluted by setting its entry in the $flag$ array to true, and the coverage counts of nodes in these polluted mRR-sets are decremented to remove their contribution in round i. The $flag$ array is reused in Section 3.5 for seed selection. Thus, we defer the details of `Discover` to there and focus here on updating mRR-sets.

Theoretical Analysis on Updating a Clean mRR-set. Let R be a clean mRR-set from round $i-1$ with root set K of size k_{i-1}. Because R contains no node in A_{i-1}, its reverse BFS and root distribution remain valid on G_i. Since $k_i \geq k_{i-1}$, we update R by keeping all existing nodes and appending $k_i - k_{i-1}$ additional roots and their reverse BFS expansions to meet the root requirement in round i.

Specifically, in Lemma 3.1, we show that when adding $k_i - k_{i-1}$ new roots, instead of sampling them all from the residual graph G_i, we can split the new roots into two subsets. First, we can sample k_∇ nodes uniformly from $R \backslash K$ and promote them to roots. Since these promoted roots already exist in R, they do not require additional reverse BFS, saving computation. The remaining $k_\Delta = k_i - k_{i-1} - k_\nabla$ new roots are sampled uniformly at random from $V_i \backslash R$ in G_i and expanded via reverse BFS by the `Probe` kernel. Lemma 3.1 establishes the theoretical correctness of this update strategy. All proofs in this paper are in the technical report [2].

Lemma 3.1. *If the previous mRR-set R with k_{i-1} roots K is clean, the nodes in R can be preserved. Additionally, we can select $k_\Delta = (k_i - k_{i-1}) \cdot \frac{n_i - |R|}{n_i - k_{i-1}}$ roots from $V_i \backslash R$ and the rest $k_\nabla = k_i - k_{i-1} - k_\Delta$ roots from $R \backslash K$ uniformly at random. The resultant root set has the same distribution as one directly generated from V_i.*

Theoretical Analysis on Updating a Polluted mRR-set. For a polluted mRR-set R from round $i-1$, since it contains at least one node in A_{i-1}, its reverse BFS generated on G_{i-1} is no longer



(a) The residual graph G_2 (b) Root sets

Figure 3: A residual graph and root sets (activated nodes are struck through).

valid on G_i. A naive remedy is to discard all polluted mRR-sets and regenerate them by sampling fresh roots from G_i. However, this alters the global distribution of root sets over all mRR-sets and thus breaks the approximation guarantee, as illustrated in Example 3.2.

Example 3.2. Assume the algorithm starts with $\eta = 4$ on graph G under the IC model in Figure 1(a). We sample sufficiently many mRR-sets, each with $\lceil n/\eta \rceil = 2$ roots. Figure 3(b) enumerates all $\binom{6}{2} = 15$ possible root sets, each with probability $1/15$. In the first round, v_2 is selected and, under the realization in Figure 1(b), activates v_2 and v_6. In the next round, we work on the residual graph G_2 in Figure 3(a) with $\eta_2 = 2$, still requiring mRR-sets with $\lceil n_2/\eta_2 \rceil = 2$ roots. If, during update, we regenerate every polluted mRR-set by resampling both roots uniformly from the remaining nodes, the root distribution becomes biased. For instance, $\frac{3}{5}$ of the mRR-sets whose root set contains v_2 or v_6 are regenerated, while mRR-sets such as $\{v_1, v_3\}$ never include v_2. In total, $\frac{11}{15}$ mRR-sets are regenerated, and the final probability of sampling root set $\{v_1, v_3\}$ becomes $\frac{1}{15} + \frac{11}{15} \cdot \frac{1}{6} = \frac{17}{90}$, deviating from the uniform probability $\frac{1}{6}$ and thus introducing sampling bias.

To address this, Lemma 3.3 shows that we should retain all non-activated roots in polluted mRR-sets and resample only the activated ones. For a polluted mRR-set R with root set K of size k_{i-1}, let $K_1 = K \setminus A_{i-1}$ and $K_2 = K \cap A_{i-1}$. We discard the roots in K_2 and sample $|K_2| + k_i - k_{i-1}$ new roots uniformly at random from $V_i \setminus K_1$; by Lemma 3.3, the resulting root set has the same distribution as one generated directly from V_i. We then invoke the `Probe` kernel to perform reverse BFS from these roots to update the mRR-set.

Lemma 3.3. *Let K be the set of roots of a random mRR-set R from round $i-1$. We write $K = K_1 \cup K_2$, where K_1 is not activated while K_2 is. Then, in round i, we delete K_2 from K and select $|K_2| + k_i - k_{i-1}$ nodes from $V_i \setminus K_1$ uniformly at random. The distribution of the new root set is the same as the one generated from V_i directly.*

Example 3.4. To illustrate Lemma 3.3, we revisit Example 3.2. We keep all clean mRR-sets and retain non-activated roots in polluted ones. For root set $\{v_1, v_3\}$, $\frac{2}{15}$ of mRR-sets already contain v_1 or v_3. The resampling of the other root results in $\{v_1, v_3\}$ with probability $\frac{1}{3}$, contributing $\frac{2}{15} \cdot 2 \cdot \frac{1}{3}$. Among the $\frac{1}{15}$ mRR-sets with roots $\{v_2, v_4\}$, regeneration yields $\{v_1, v_3\}$ with probability $\frac{1}{6}$, contributing $\frac{1}{15} \cdot \frac{1}{6}$. The original $\frac{1}{15}$ mRR-sets with roots $\{v_1, v_3\}$ remain unchanged. Thus the final probability of $\{v_1, v_3\}$ is $\frac{2}{15} \cdot 2 \cdot \frac{1}{3} + \frac{1}{15} \cdot \frac{1}{6} + \frac{1}{15} = \frac{1}{6}$. A symmetric argument shows every root set has probability $\frac{1}{6}$, confirming unbiased root-set sampling.

Circular Segment Update of mRR-sets on `GmRR`. With the theoretical analysis, we now describe how to efficiently update mRR-sets within a `GmRR` segment. Recall that each thread block exclusively

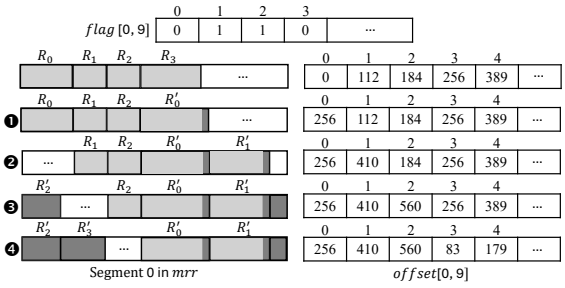


Figure 4: Circular segment update of a segment in `GmRR`.

owns one segment and is responsible for generating and updating its mRR-sets. The top-left part of Figure 4 illustrates Segment 0 with four mRR-sets $R_{0,1,2,3}$ generated by Block 0 in round $i-1$, where R_0 and R_3 are clean, while R_1 and R_2 are polluted as indicated in the $flag$ array. In round i, all four mRR-sets must be updated.

A straightforward approach is to update mRR-sets sequentially from R_0 to R_3. Since R_0 is clean, we add new roots and perform reverse BFS, increasing its size and forcing us to shift subsequent mRR-sets R_1–R_3 backward to free space for the expanded R_0. Updating the polluted mRR-set R_1 likewise changes its size after root resampling and reverse BFS, so all subsequent mRR-sets must be shifted again. Repeating this process for all mRR-sets requires moving each set multiple times, which is inefficient when many mRR-sets reside in a segment, and the unknown final size of each updated mRR-set also requires temporary GPU buffers.

Instead, we design a circular segment update strategy that moves each mRR-set at most once, and writes new content circularly within the segment, so no subsequent mRR-sets are shifted. Steps ❶–❹ in Figure 4, together with the $offset$ array, illustrate this process. As explained in Section 3.2, each `GmRR` segment stores mRR-sets sequentially in mrr, with $offset$ recording the start of their positions and the free space. Especially, we first update the last mRR-set in the segment in place. In Figure 4, the last mRR-set R_3 is updated to R'_0, which becomes the first mRR-set in the segment for round i (Step ❶); its light-gray region is the reused content, and its dark-gray region is the part newly probed from added roots. Accordingly, we set $offset[0] = 256$, the start position of R'_0, which also indicates that R'_0 is now the first mRR-set in the segment.

Next, we update R_0 from round $i-1$ to R'_1. Instead of modifying R_0 in place (which would require shifting subsequent mRR-sets), we write R'_1 immediately after R'_0, leaving the original space of R_0 free (white in Step ❷). Originally, R_0 is the first mRR-set; after the update it becomes the second, at $offset[1] = 410$.

We then update R_1 to R'_2 and place it after R'_1, where R'_2 has start position $offset[2] = 560$, making it the third mRR-set in the segment (Step ❸). When the segment end is reached, the remaining free space lies at its beginning. Using circular addressing within the segment (mapping each physical index modulo the segment length ℓ), the portion of R'_2 that exceeds the segment end is wrapped to the segment front (dark gray in Step ❸). The free space after R'_2 is then used to update R_2 into R'_3 with start position $offset[3] = 83$; since it is the last mRR-set, its end position is stored at $offset[4] = 179$, which also indicates the start of the free space (Step ❹).

The `ParallelUpdate` Kernel. Algorithm 4 presents the kernel `ParallelUpdate`, which updates mRR-sets in each segment by a thread block. The input is the θ_{i-1} mRR-sets in `GmRR` from round

$i-1$, represented by mrr and $offset$, and the array $flag$ marking each mRR-set as polluted or clean. For each block, we first compute its local mRR-set count r (Line 1), set r_1 to index the updated mRR-sets and r_2 to index the mRR-set being processed (Line 2), where $r_2 = r-1$ so that we start update from the last mRR-set. ℓ_b is the base address of the block's segment. We then read the $start$ and end of the r_2-th mRR-set and set b_w to $start$ as the write position for its updated version (Lines 3–4). In each iteration (Line 5), we mark all nodes in V_i as unvisited in parallel, then set both b_r and b_w to the new start position of the mRR-set pending update (Line 6), and compute its original length len under the circular layout (Line 7).

From Lines 8 to 18, if the r_2-th mRR-set is clean, we first obtain k_Δ and k_∇ according to Lemma 3.1 (Line 9). We then retain all original nodes of this mRR-set by copying them to their new position and marking them as visited in the updated mRR-set (Lines 10–12), with all operations executed in parallel. Next, we sample k_Δ new roots and write each new root to position $(b_w+len+j)\%\ell + \ell_b$ of mrr, updating the $visited$ and cnt arrays accordingly (Lines 13–15). We then perform reverse BFS from these k_Δ new roots by invoking the `Probe` kernel (Line 17). Note that the k_Δ new roots and k_∇ promoted roots are not necessarily contiguous with the original roots of this mRR-set. As described in Section 3.2, each mRR-set in mrr stores all roots first, followed by nodes probed by reverse BFS. Thus, in Line 18 we swap the positions of the k_Δ and k_∇ roots so that the first k_i entries of the updated mRR-set are its roots.

From Lines 20 to 26, if the r_2-th mRR-set is polluted, we first retrieve its previous roots at their previous positions in the existing mRR-set (Lines 21–22) and, for each root u, test whether $u \in V_i$. If $u \notin V_i$ (i.e., it was activated in the previous round) or fewer than k_i valid roots remain, we sample replacement roots so that the updated mRR-set still has exactly k_i roots (Lines 24–25). Finally, we write all roots to their new positions in mrr and atomically increase the coverage of each root by 1 (Line 26). Then `Probe` is invoked from these k_i roots to get the mRR-set (Line 28). Once the r_2-th mRR-set is updated, we move to the next one by updating r_2, recomputing its original $start$ and end, recording the start position of the updated mRR-set in $offset$, and incrementing r_1 (Lines 29–31). After all mRR-sets are processed, we set b_w as end of last updated mRR-set, which is also the start of the free space in the segment (Line 32).

Update Workload Balancing. In `ParallelUpdate`, the main cost comes from updating polluted mRR-sets, which requires more BFS traversals than clean ones. Thus, the workload per segment in `GmRR` is largely determined by its number of polluted mRR-sets. This number can be highly skewed across segments, causing severe inter-block imbalance and poor GPU utilization. To alleviate this, we adopt a lightweight load-balancing scheme. Using the $flag$ array, we first count in parallel the polluted mRR-sets in each segment, then sort all segment IDs in decreasing order of this count and partition them into β groups, each containing as many segments as blocks. Segments in the same group have similar workloads, and `ParallelUpdate` processes groups one by one over β iterations, which reduces workload imbalance among blocks.

Algorithm 4: The `ParallelUpdate` Kernel

Input: Graph G_i, cnt, mrr, $offset$, $flag$, θ_{i-1}.

1 $r \leftarrow \theta_{i-1}/gridDim.x + \mathbb{I}_{\{(\theta_{i-1}\ \%\ gridDim.x) > block.id\}}$;
2 $r_1 \leftarrow 0$; $r_2 \leftarrow r-1$; $\ell_b \leftarrow block.id * \ell$;
3 $start \leftarrow offset[r_2 + block.id * L]$; $b_w \leftarrow start$;
4 $end \leftarrow offset[r_2 + 1 + block.id * L]$;
5 **while** $r_1 < r$ **do**
6 $\quad \forall u \in V_i, visited[u] \leftarrow false$ in **parallel**; $b_r \leftarrow b_w$;
7 $\quad len \leftarrow (end > start)\ ?\ (end - start) : \ell - (start - end)$;
8 $\quad$ **if** $flag[r_2 + block.id * L] == false$ **then** `// clean`
9 $\quad\quad$ Obtain k_Δ and k_∇ *w.r.t.* Lemma 3.1;
10 $\quad\quad$ **parallel for** $j = thread.id$; $j < len$; $j \mathrel{+}= blockDim.x$ **do**
11 $\quad\quad\quad u \leftarrow mrr[(start + j)\ \%\ \ell + \ell_b]$; $visited[u] \leftarrow true$;
12 $\quad\quad\quad mrr[(b_w + j)\ \%\ \ell + \ell_b] \leftarrow u$;
13 $\quad\quad$ **parallel for** $j = thread.id$; $j < k_\Delta$; $j \mathrel{+}= blockDim.x$ **do**
14 $\quad\quad\quad$ Atomically sample $u \in V_i$ with $visited[u] == flase$ and set $visited[u] \leftarrow true$;
15 $\quad\quad\quad mrr[(b_w+len+j)\%\ell+\ell_b] \leftarrow u$; atomicAdd($cnt[u]$, 1);
16 $\quad\quad$ __syncthreads();
17 $\quad\quad$ `Probe`(G_i, k_Δ, $b_r + len$, b_w, mrr, $visited$, cnt);
18 $\quad\quad$ **parallel** swap k_Δ and k_∇ roots to after previous k_{i-1} roots;
19 $\quad$ **else** `// polluted`
20 $\quad\quad$ **parallel for** $j = thread.id$; $j < k_i$; $j \mathrel{+}= blockDim.x$ **do**
21 $\quad\quad\quad$ **if** $j < k_{i-1}$ **then**
22 $\quad\quad\quad\quad u \leftarrow mrr[(start+j)\%\ell+\ell_b]$; $visited[u] \leftarrow true$;
23 $\quad\quad\quad$ __syncthreads();
24 $\quad\quad\quad$ **if** $u \notin V_i$ *or* $j \geq k_{i-1}$ **then**
25 $\quad\quad\quad\quad$ Atomically sample $u \in V_i$ with $visited[u] == flase$ and set $visited[u] \leftarrow true$;
26 $\quad\quad\quad mrr[(b_w+j)\%\ell+\ell_b] \leftarrow u$; atomicAdd($cnt[u]$, 1);
27 $\quad\quad$ __syncthreads();
28 $\quad\quad$ `Probe`(G_i, k_i, b_r, b_w, mrr, $visited$, cnt);
29 $\quad r_2 \leftarrow (r_2 + 1)\ \%\ r$; $start \leftarrow offset[r_2 + block.id * L]$;
30 $\quad end \leftarrow offset[r_2 + 1 + block.id * L]$;
31 $\quad offset[r_1 + block.id * L] \leftarrow b_r$; $r_1 \leftarrow r_1 + 1$;
32 $offset[r_1 + block.id * L] \leftarrow b_w$;

3.5 Parallel Seed Selection

In round i with θ_i mRR-sets, we greedily select a batch S_i of b seeds that maximizes the coverage-to-cost ratio. If S_i satisfies the guarantee, we accept it; otherwise, we generate additional mRR-sets and repeat the selection. When a node u is selected, all mRR-sets containing u are marked as covered and excluded from subsequent coverage computations with coverage counts updated.

To this end, we use the GPU kernel `Select` (Algorithm 5) to perform one seed-selection attempt and obtain S_i. We initialize $S_i = \emptyset$ with $\Lambda(S_i) = 0$, copy cnt to the working array cnt' (since this attempt may be rejected), and set all entries of $flag$ to false, as no mRR-set is covered before the first selection. In each iteration (Lines 2–5), we perform a GPU parallel reduction to select the node $u \in G_i$ with the maximum coverage-to-cost ratio, add u to S_i, and update $\Lambda(S_i)$. We then invoke the `Discover` kernel (Algorithm 6) to mark mRR-sets containing u as covered and decrement the corresponding coverage counts cnt'. After b seeds are selected, we test if S_i is qualified by computing the lower bound $\Lambda^l(S_i)$ of its coverage and the upper bound $\Lambda^u(S_i^o)$ for the optimal seed batch. If $\Lambda^l(S_i) > \rho_b(1-\hat{\epsilon})\Lambda^u(S_i^o)$ holds, S_i is accepted, we overwrite cnt with cnt', and return S_i (Line 7); otherwise, we return an empty set (Line 8), indicating to `GAAS` that additional mRR-sets are needed.

Algorithm 5: The `Select` Kernel

Input: Batch size b, cnt, mrr, $offset$, $flag$, θ_i.

$S_i \leftarrow \emptyset; \Lambda(S_i) \leftarrow 0; flag[\cdot] \leftarrow false; cnt'[\cdot] \leftarrow cnt[\cdot]$;
while $|S_i| < b$ **do**
  $u \leftarrow \text{argmax}_{v \in V_i} \frac{cnt'[v]}{c(v)}$ by **parallel** reduction on GPU;
  $S_i \leftarrow S_i \cup \{u\}, \Lambda(S_i) \leftarrow \Lambda(S_i) + cnt'[u]$;
  Parallel `Discover`(cnt', mrr, $offset$, $flag$, θ_i, $\{u\}$);
Derive $\Lambda^l(S_i)$ and $\Lambda^u(S_i^o)$;
if $\Lambda^l(S_i) > \rho_b(1-\hat{\epsilon})\Lambda^u(S_i^o)$ **then** $\{cnt[\cdot] \leftarrow cnt'[\cdot]$; **return** S_i;$\}$
else return $\emptyset$;

Algorithm 6: The `Discover` Kernel

Input: cnt, mrr, $offset$, $flag$, θ_i, node set A.

1 $r \leftarrow \theta_i / gridDim.x + \mathbb{I}_{\{(\theta_i \,\%\, gridDim.x) > block.id\}}$; $r_1 \leftarrow 0$;
2 **while** $r_1 < r$ **do**
3 $start \leftarrow offset[r_1 + block.id * L]$;
4 $end \leftarrow offset[r_1 + 1 + block.id * L]$;
5 $len \leftarrow (end > start)\ ?\ (end - start) : \ell - (start - end)$;
6 **if** $flag[r_1 + block.id * L] == false$ **then**
7 **parallel for** $j = thread.id; j < len; j \mathrel{+}= blockDim.x$ **do**
8 **if** $mrr[(start + j)\,\%\,\ell + block.id * \ell] \in A$ **then**
9 $flag[r_1 + block.id * L] \leftarrow true$;
10 **if** $flag[r_1 + block.id * L] == true$ **then break**;
11 __syncthreads();
12 **parallel for** $j = thread.id; j < len; j \mathrel{+}= blockDim.x$ **do**
13 $u \leftarrow mrr[(start + j)\,\%\,\ell + block.id * \ell]$;
14 atomicSub($cnt[u]$, 1);
15 __syncthreads();
16 $r_1 \leftarrow r_1 + 1$;

In Algorithm 6, given a node set A (equal to $\{u\}$ in the `Select` kernel), `Discover` identifies all mRR-sets covered by A and updates the coverage counts of the nodes in those mRR-sets. In Section 3.4, we have also used `Discover` to identify the mRR-sets covered, i.e., polluted by the activated nodes in A_{i-1}. In each thread block, we first obtain the number of mRR-sets r in its segment (Line 1), then iterate over them with r_1 from 0 to $r-1$ (Lines 2–16). For each mRR-set indexed by r_1, we first obtain its start and end positions from the $offset$ array and its length using the `GmRR` layout in Section 3.2 (Lines 3–5). If its flag is false (Line 6), meaning it is not yet covered, all threads in the block (Line 7) scan this mRR-set in parallel to check whether any node is in A (Line 8); once such a node is found, they set the flag to true and terminate the scan (Lines 9–10). All threads in the block then collaboratively scan the nodes in this mRR-set and decrement coverage counts in cnt (Lines 12–14). The loop repeats until all mRR-sets in the segment have been processed.

3.6 The Complete GAAS Algorithm

Algorithm 7 describes the complete workflow of `GAAS`. Given the graph G, node costs $c(\cdot)$, influence target η, estimation error ϵ, and batch size b (seeds per round), we initialize the round index $i = 1$, the residual graph G_i, the residual target η_i, the seed set S, and allocate `GmRR` on the GPU with arrays mrr, $offset$, $flag$, and cnt.

While $\eta_i > 0$, we first compute the required number of mRR-sets θ_i for round i and its upper bound $\theta_{\max}$. If mRR-sets from the previous round exist (i.e., $\theta_{i-1} > 0$), we update and reuse them

Algorithm 7: `GAAS`

Input: Graph G, target η, estimation error ϵ, batch size b, cost $c(\cdot)$.
Output: Seed set S.

1 $i \leftarrow 1, G_i \leftarrow G, \eta_i \leftarrow \eta, \rho_b \leftarrow 1 - (1 - 1/b)^b, \theta_0 \leftarrow 0, S \leftarrow \emptyset$;
2 Allocate and initialize arrays mrr, $offset$, $flag$, and cnt on GPU;
3 **while** $\eta_i > 0$ **do**
4 Get θ_i and $\theta_{\max}$;
5 **if** $\theta_{i-1} > 0$ **then**
6 `ParallelUpdate`(G_i, cnt, mrr, $offset$, θ_{i-1}, $flag$);
7 $S_i \leftarrow$ Kernel `Select`(b, cnt, mrr, $offset$, θ_{i-1}, $flag$);
8 **if** $S_i \neq \emptyset$ **then** $\{\ \theta_i \leftarrow \theta_{i-1}$; **goto** Line 15; $\}$
9 **else** $\theta_i \leftarrow 2\theta_{i-1}$;
10 **while** $\theta_i < \theta_{\max}$ **do**
11 `ParallelGen`(G_i, cnt, mrr, $offset$, θ_{i-1}, θ_i);
12 $S_i \leftarrow$ Kernel `Select`(b, cnt, mrr, $offset$, θ_i, $flag$);
13 **if** $S_i \neq \emptyset$ **then break**;
14 $\theta_{i-1} \leftarrow \theta_i$; $\theta_i \leftarrow 2\theta_i$;
15 Diffuse from S_i, and obtain the activated nodes A_i;
16 Kernel `Discover`(cnt, mrr, $offset$, $flag$, θ_i, A_i);
17 $S \leftarrow S \cup S_i; G_{i+1} \leftarrow G_i \backslash A_i; \eta_{i+1} \leftarrow \eta_i - |A_i|; i \leftarrow i + 1$;

for seed selection using the designs in Sections 3.4 and 3.5: we call `ParallelUpdate` to update all existing mRR-sets in parallel (Line 6), and then invoke `Select` to obtain b seeds S_i (Line 7). If S_i is non-empty (Line 8), the current mRR-sets suffice and we proceed directly to Line 15 without further generation; otherwise, we set $\theta_i \leftarrow 2\theta_{i-1}$ (Line 9) and continue with the while loop. In this loop, we progressively enlarge the mRR-set collection until either $S_i \neq \emptyset$ or θ_i reaches $\theta_{\max}$. In each iteration, we invoke `ParallelGen` to get additional mRR-sets (Line 11), then run `Select` on the enlarged collection (Line 12); if S_i remains empty, we double θ_i and repeat.

Once a non-empty batch S_i is obtained, we simulate diffusion from S_i to obtain the activated nodes A_i (Line 15). We then call `Discover` (Line 16) to mark polluted mRR-sets in $flag$ for the next round and to decrement cnt so their contributions are removed. Finally, we update $S \leftarrow S \cup S_i$, construct the new residual graph G_{i+1}, derive the new target η_{i+1} (Line 17), and proceed to the next round until the overall influence target is met.

In practice, the required number of mRR-sets usually decreases over adaptive rounds, so we periodically clear all segments and regenerate mRR-sets for the current round. This occasional regeneration discards excess mRR-sets, and its cost is amortized by the computational savings from updates in most rounds.

The theoretical guarantee of `GAAS` is given in Theorem 3.5, where π^o is the optimal policy that incurs the same per-round seed cost as `GAAS`. The approximation factor scales linearly with $\ln \eta$, improving the guarantees of [14, 39] by a factor of $\ln \eta$.

Theorem 3.5. *The cost of* `GAAS` *reaching* η *approximates that of* π^o *in the form:* $\mathbb{E}[c(\texttt{GAAS})] \leq \frac{2 \ln \eta}{(1-\epsilon)(1-1/e)\rho_b} \mathbb{E}[c(\pi^o)] + b + 1$.

4 RELATED WORK

Minimum Cost Seed Selection (MCSS). The MCSS problem aims to find a seed set that minimizes the cost to achieve the influence threshold η. First formulated in [30], the problem shares the submodularity and monotonicity properties with the classic Influence Maximization (IM) problem. Existing approaches to MCSS can be

divided into two categories. The first category focuses on reaching η only in expectation [12, 15, 18, 30], where selected seeds can only guarantee the coverage meets η. However, the actual influence spread of the selected seed often falls significantly short of the target η. Therefore, they are not feasible for AMCSS that requires the selected nodes must influence at least η nodes in every realization. The other category addresses this practical limitation by providing probabilistic guarantees [12, 51], ensuring that with a predefined probability p the influence achieved by the selected seeds reaches η in a random realization. Thus, we can set a high probability value to increase the likelihood that the selected seeds meet the target influence threshold in practice.

Adaptive Seed Minimization (ASM). The adaptive setting in IM and MCSS receives considerable attention in recent years [6, 9, 13, 14, 16, 19, 22, 35, 39, 43, 44, 52]. In the adaptive setting, nodes are seeded in batches at different rounds, and after each batch is seeded, an actual diffusion process is allowed to unfold, with the outcomes observed before deciding the next batch of seeds. Golovin and Krause [13] pioneered the extension of the classic notions of submodularity and monotonicity to the adaptive setting, thereby establishing an important theoretical foundation. However, their work did not provide a practical algorithm tailored for the ASM problem. Tang et al. [39] first introduced truncated influence, recognizing that influence beyond a threshold η provides no additional value, and they devised mRR-sets to estimate marginal truncated influence targeted for ASM. However, ASTI assumes uniform costs for all nodes and overlooks the fact that the cost of nodes are heterogeneous in real-world influence campaigns. Moreover, ASTI is a single-threaded CPU-based implementation and regenerates the entire collection of mRR-sets in each round.

Parallel RR-set Approaches. Several methods have been proposed to solve the IM problem in parallel [31, 32, 37] using the Reverse Reachable set (RR-set), which uses only one root. While the RR-set generation phase can be adapted for mRR-set generation in AMCSS, these approaches face notable efficiency challenges. Ripples and cuRipples [31, 32] utilize thread-level parallelization for RR-set generation. However, these methods suffer from workload imbalance due to skewed RR-set sizes and suboptimal thread utilization when RR-sets are fewer than the available threads. Additionally, each thread requires its own auxiliary arrays (*e.g.*, for tracking visited nodes), limiting scalability to large graphs. These issues remain when it is extended to mRR-set generation in adaptive rounds. For the block-level parallelization approach [37], its implementation necessitates temporary storage and frequent atomic operations to manage RR-sets in global memory, also degrading performance and introducing further memory constraints.

5 EXPERIMENTS

We evaluate GAAS and baselines on a server with an Intel Xeon Gold 6226R 2.90GHz CPU with 16 physical cores, 384 GB RAM, and an NVIDIA RTX 3090 GPU with 24 GB of memory. The CPU and GPU have similar prices: approximately $1,517 and $1,499, respectively.

5.1 Experimental Setup

Datasets. We adopt four datasets used in previous work [12, 19, 22, 39, 46]. The statistics are in Table 1. These datasets represent diverse social networks from real world and are available at [27]. The large dataset, LiveJournal, contains over 4.85 million nodes and 68.5 million edges, which is the largest network evaluated in [39].

Table 1: Datasets information. ($K = 10^3, M = 10^6$)

Name	n	m	Avg. deg	Type
DBLP	317K	1.05M	6.62	undirected
YouTube	1.13M	2.99M	5.27	undirected
Flickr	2.30M	33.1M	14.4	directed
LiveJournal	4.85M	68.5M	14.1	directed

Baselines. ASTI originally handles uniform seed cost, and is single-threaded on CPU [39]. We extend it to handle non-uniform costs by selecting seeds with the largest influence-to-cost ratios in each round. We also implement a CPU-parallel version, PASTI-CPU, parallelizing all applicable computations in ASTI using OpenMP [8] utilizing all CPU cores. Moreover, we include a GPU-parallel ASTI version, PASTI-GPU, which adapts the GPU-based techniques from [37] (originally designed for single-root RR-set generation in Influence Maximization) to support fresh mRR-set generation across adaptive rounds for AMCSS. Following [37], PASTI-GPU maintains mRR-sets in a contiguous way in the global GPU memory. Finally, we compare with SCORE, a state-of-the-art non-adaptive MCSS method that provides a probabilistic guarantee on target influence threshold η in any realization. We configure SCORE with high confidence ($p = 0.99$) to align with the stopping criterion of adaptive methods which is reaching η with probability 1.0.

Parameters. We use consistent parameter settings across all tested baselines and our method GAAS. Following [12, 19, 22, 39], we set the mRR-set estimation error $\epsilon = 0.5$ and failure probability $\delta = 1/n_i$ per round. The diffusion probability of any edge (u, v) is $p(u, v) = 1/|N_{\text{in}}(v)|$, and the cost of a node u is $c(u) = c_0 + 0.01 \cdot |N_{\text{out}}(u)|$ with $c_0 = 0.01$ to avoid zero-cost nodes. We evaluate all methods under both the IC and LT models, averaging results over 10 shared random realizations. Under the IC model, we select 4 nodes per batch on DBLP and 16 nodes per batch on larger datasets to ensure the single-threaded baseline ASTI completes within two days. The influence threshold η ranges from 1K to 5K on DBLP and YouTube, from 2K to 10K on Flickr, and from 5K to 25K on LiveJournal, reflecting their different scales. Under the LT model, we fix the batch size $b = 4$ for all datasets, and vary the influence threshold η ranges according to the dataset size. Specifically, the ranges of η vary from 1K to 3K on DBLP and YouTube, from 2K to 6K on Flickr, and from 2.5K to 12.5K on LiveJournal. In GAAS, when updating mRR-sets, we set the number of groups to $\beta = 8$ for all datasets under both the IC and LT models for load balancing, as determined by our ablation study in Section 5.4. We configure GAAS and all parallel baselines to fully utilize all available CPU cores and GPU resources, respectively.

5.2 Seed Cost and Efficiency under the IC model

AMCSS aims to minimize seed cost while reaching the target influence threshold η. Thus, efficiency is comparable only when methods reliably attain η with low seed cost. By the theoretical guarantees of GAAS and the adaptive baselines, all adaptive methods always

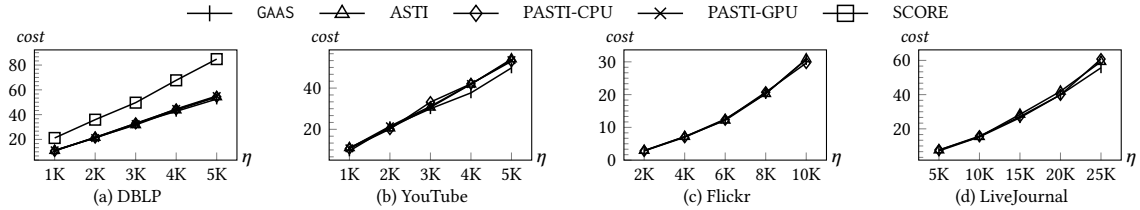


Figure 5: Seed cost of algorithms under the IC model.

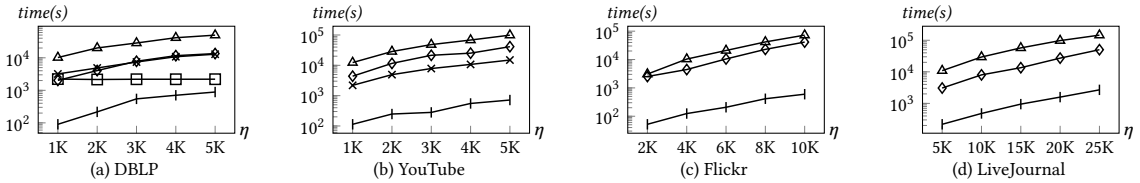


Figure 6: Running time in seconds of algorithms under the IC model.

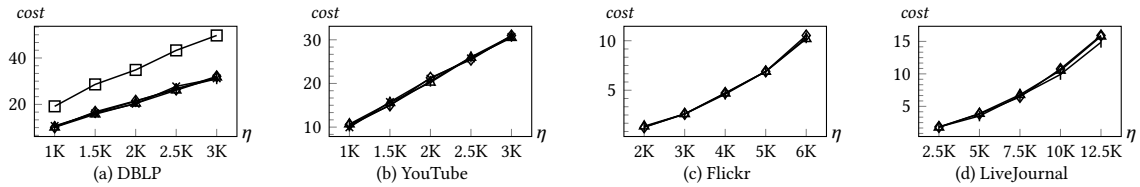


Figure 7: Seed cost of algorithms under the LT model.

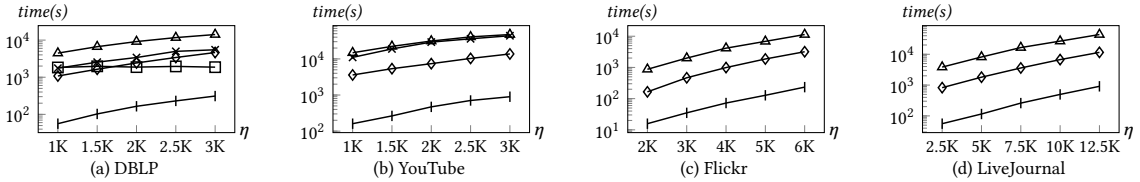


Figure 8: Running time in seconds of algorithms under the LT model.

attain η (100% *attainment*). We therefore omit attainment plots (all equal to 1.0) and focus on comparing the *seed cost* and *running time*.

Seed Cost. Figure 5 reports the seed cost each method incurs to reach various η under the IC model. All adaptive methods yield similar low costs and require substantially less seed cost than the non-adaptive method SCORE on DBLP. Because GAAS and the adaptive ASTI baselines all leverage the mRR-set estimation framework, they have similar estimation accuracy and thus comparable seed cost. Nonetheless, we observe that GAAS achieves slightly lower cost than other methods, especially on YouTube and LiveJournal. This is because, when updating mRR-sets in adaptive rounds, GAAS deliberately retains all previous mRR-sets, even though the number of required mRR-sets typically decreases over rounds, allowing it to exploit more mRR-sets than regenerating-based methods and thus make higher-quality seeding decisions. The higher cost of SCORE arises from its non-adaptive nature: it cannot leverage intermediate results during seeding and therefore selects more nodes to ensure reaching η with the required probability. Moreover, SCORE runs out of memory (OOM) on the other three larger datasets, since it depends on graph sketches to provide the probabilistic guarantee, and the sketch size grows rapidly as the graph scales.

Running Time. Given the minimal and similar seed costs, we next compare the efficiency of different methods. Figure 6 reports the running time (y-axis in log scale). The results show that GAAS consistently runs faster than all baselines, often by more than an order of magnitude. For all methods, running time increases with larger η since higher thresholds require more rounds to select enough seeds. Compared with the strongest GPU-based baseline PASTI-GPU, GAAS achieves speedups of up to 33.9× and 24.2× on DBLP and YouTube, respectively. These gains stem from our GmRR structure, which enables in-segment mRR-set generation and update directly on the GPU. In contrast, PASTI-GPU runs out of memory on the larger Flickr and LiveJournal graphs for two main reasons. First, it relies heavily on dynamic memory allocation, which leads to severe memory fragmentation. Second, it temporarily buffers mRR-sets being generated in global memory, effectively doubling peak memory usage in the worst case. GAAS avoids these issues by combining the GmRR structure with carefully designed parallel mRR-set generation and update kernels, which use global memory and thread blocks more efficiently on the GPU. On the two largest datasets, GAAS achieves substantial speedups over the strongest CPU baseline PASTI-CPU. On Flickr, GAAS runs up to 68.9× faster than PASTI-CPU when η = 10K, and on LiveJournal it consistently attains more than 10× speedup across all evaluated settings. For SCORE on DBLP, its running time in Figure 6(a) is less informative because it incurs significantly higher seed cost, as shown in Figure 5(a). Even so, GAAS remains much faster than SCORE on DBLP with lower seed cost. SCORE runs OOM on other datasets.

5.3 Seed Cost and Efficiency under the LT model

Under the LT model, GAAS is also faster than existing methods, often by an order of magnitude, while maintaining low seed cost.

Seed Cost. Figure 7 reports the seed cost under the LT model. The trends are consistent with those in Figure 5. For each dataset and influence threshold η, all adaptive methods incur similar costs, whereas the non-adaptive SCORE requires substantially higher cost on DBLP and runs out of memory on larger datasets. Overall, GAAS achieves comparable or lower cost than all baselines, with its advantage increasing for larger η, where additional rounds allow it to reuse and update existing mRR-sets instead of regenerating them, leading to more accurate truncated influence estimation.

Time Efficiency. Figure 8 reports the running time under the LT model. The results show that GAAS consistently outperforms all baseline methods across all evaluated datasets. Note that SCORE and PASTI-GPU run OOM on the two largest datasets and thus do not produce results there. Across all influence targets η where the baselines succeed, GAAS achieves more than 10× speedup over the strongest competitor. In particular, GAAS attains speedups of up to 19.2× on DBLP, 22.3× on YouTube, 14.6× on Flickr, and 15.7× on LiveJournal compared to the fastest baseline. These substantial performance gains arise from our carefully designed GPU optimizations, including independent inter-block mRR-set generation and the efficient selective reuse of clean mRR-sets, which together minimize redundant computation and fully exploit GPU parallelism.

5.4 Experimental Analysis

Time Breakdown and the Effect of `ParallelUpdate`. A key design in GAAS is to update and reuse mRR-sets from previous rounds instead of regenerating them from scratch by the GPU kernel `ParallelUpdate`. To quantify the efficiency gain, we compare GAAS with a variant GAAS_{gen} that does not perform mRR-set updates and instead regenerates required mRR-sets from scratch in every round on GPU. For each method, we decompose the running time into three components: mRR-set generation, mRR-set update, and others (including seed selection). In Figure 9, on each dataset, we report the time breakdown of GAAS and GAAS_{gen} under the IC model. Under the same η, the bar with ⋆ represents GAAS, while the bar without ⋆ represents GAAS_{gen}. The results show that our parallel update mechanism with the `ParallelUpdate` kernel significantly reduces time spent on mRR-set generation: in GAAS_{gen}, mRR-set generation dominates the runtime, whereas in GAAS it becomes negligible and the total running time is much lower. This confirms that GAAS effectively reuses existing mRR-sets by updating mRR-sets in parallel on the GPU. The results under the LT model are deferred to the technical report [2] and are similar to the IC model. Overall, GAAS consistently outperforms GAAS_{gen} in efficiency, confirming the effectiveness of our proposed mRR-set update kernel.

GPU Profiling. We use NVIDIA Nsight Compute [1] to quantify the efficiency of our GPU method via standard GPU performance metrics. Figure 10 reports the GPU occupancy level, instructions executed per cycle (IPC), and streaming multiprocessor (SM) active percentage—where higher values indicate better performance—as well as the number of atomic operations, where lower values are preferred for efficiency, comparing GAAS with the PASTI-GPU baseline on DBLP and YouTube under the IC model. Clearly, GAAS achieves

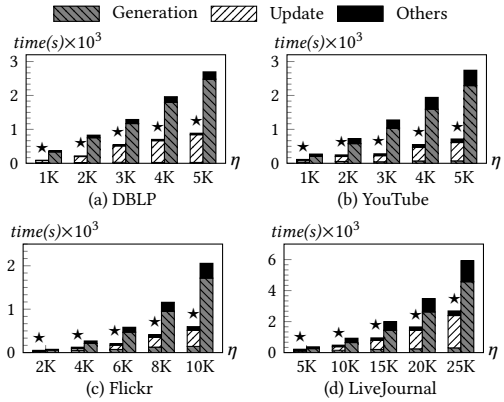


Figure 9: Runtime breakdown under the IC model; for each η, the left bar with ⋆ is GAAS and the right bar is GAAS_{gen}.

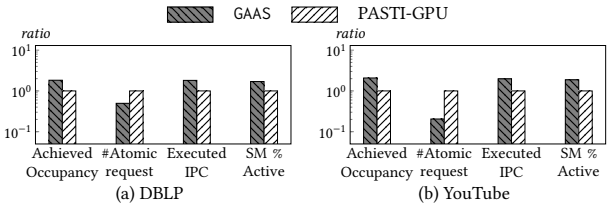


Figure 10: GPU profiling analysis relative to PASTI-GPU.

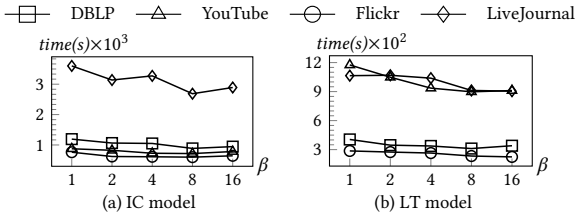


Figure 11: Vary β under largest tested η across datasets.

better GPU occupancy, higher IPC, and higher SM throughput while requiring fewer atomic operations than PASTI-GPU. For example, GAAS attains 1.8× and 2.1× higher occupancy than PASTI-GPU on these datasets, respectively. These results demonstrate that GAAS utilizes GPU resources more efficiently than PASTI-GPU, owing to our algorithmic designs and performance optimizations.

Vary β for load balancing. In the `ParallelUpdate` kernel, we employ a load-balancing strategy controlled by a parameter β, which specifies the number of groups used for mRR-set updates on the GPU. We vary $\beta \in \{1, 2, 4, 8, 16\}$ to study its impact under both the IC and LT models. When $\beta = 1$, no load balancing is applied. Figure 11 reports the effect of β under both models. We observe that the running time generally decreases as β increases from 1 to 8. However, when β increases from 8 to 16, the running time starts to increase, because the overhead of finer-grained grouping and additional block-level synchronization outweighs the marginal load-balancing benefits. Therefore, we set $\beta = 8$ in all experiments.

6 CONCLUSION

We present a GPU-accelerated AMCSS method GAAS with provable guarantees by integrating novel algorithmic ideas with GPU-aware system design. We design a GPU-tailored mRR-set structure GmRR

that divides the pre-allocated global memory into equal-size circular segments, each exclusively owned by one thread block, to boost parallelism. On top of GmRR, we build a ParallelGen kernel for mRR-set generation, a ParallelUpdate kernel that reuses and updates previous mRR-sets via theoretically sound rules and circular in-segment updates with load balancing, and a Select kernel for parallel seed selection. Experiments on real-world graphs show that GAAS is over an order of magnitude faster than the baselines, while achieving comparable or lower seed cost. As future work, we plan to extend GAAS to distributed or multi-GPU settings.

7 NOTATIONS

Table 2: Frequently used notations.

$G = (V, E)$	social network with users V and edges E
$G_i = (V_i, E_i)$	residual graph in round i
n, m	number of users and edges
$N_{\text{in}}(u)$, $N_{\text{out}}(u)$	in-/out-neighbors of u
$p(e)$, $p(u, v)$	diffusion probability of edge e or (u, v)
θ_i	# of mRR-sets used in round i
k_i	# of roots in mRR-sets in round i
ϵ	estimation error of mRR-sets
η	target influence threshold
η_i	remaining target influence in round i
S_i	seed set selected in round i
A_i	activated nodes in round i
$c(v)$	cost of seeding node v
$I(S)$	random influence spread of S
ϕ, Φ	specific / random realization
$\Gamma_i(S)$	truncated influence of S in G_i
$\hat{\Gamma}(S)$	estimated truncated influence
π	adaptive seed selection policy

8 MISSING PROOFS

8.1 Proof of Lemma 3.1

We first prove that the nodes in a clean mRR-set R can be preserved. Then, we show that our way of adding roots is correct.

(1) Note that the root distribution of an unpolluted mRR-set R is the same as selecting the roots in the new residual graph. Let the original root set of R be K. The probing result that generates R can be represented as a tree structure T. The fact that R is not polluted indicates that the frontier of the reverse BFS from K (i.e., the leaves) did not probe activated nodes or the activation failed. For each probe, the probability of its success is not affected by whether it is probed in G_{i-1} or G_i. Thus, the probability of reaching the nodes in R is the same as reaching them in G_i. Similarly, for the nodes in V_i that were probed but failed, the probability is also the same. For the failed probes w.r.t. nodes in $V_{i-1} \backslash V_i$, since they are not in V_i, they will not be in R as well. Thus, the composite of R will not change in a new generation in G_i.

Then, we show that when new roots are selected, the nodes in R can still be preserved. The selection of new seeds results in two cases. For the roots that were in R, we argue that the composite of R will not change. For each node $u \in R$, there must be a path from one of the root (e.g. r) to u. When some node v in the path is promoted to be a root. The path from r to v will not change, but the nodes after v (denoted as $O = \{o_1, o_2, \cdots, o_l\}$, where l is the length) may be traversed by some other nodes in R. We can prove that the nodes in O will still be in R, by induction. For the first node o_1, it may be probed by a new node other than v. If the probe is successful, o_1 will stay in R. Otherwise, at least o_1 can be traversed by v. Either way, o_1 will still be in R. Similarly, assuming o_i is in R, we can prove that o_{i+1} is still in R. Thus, the nodes in R will not change.

Thus, the elements in R have the same distribution as an mRR-set generated from scratch in the new residual graph.

(2) To validate our way of adding roots, we first invoke Lemma 3.3, which shows that by replacing the activated roots with randomly selected ones, the root set is equal to a newly selected root set. Then, in this lemma, we do not consider the case where there are activated roots.

Assume a random mRR-set R with k_{i-1} roots K, which needs $\Delta = k_i - k_{i-1}$ new roots. When adding new roots into R, to mimic the generation of fresh mRR-sets, we only avoid selecting the roots already in R. Then, some non-root nodes in R may be selected as a root. To accommodate this case, we select some new roots from $R \backslash K$ and some from $V_i \backslash R$. According to Alg. 4, to ensure a uniform sampling, the numbers of roots we select from $R \backslash K$ and $V_i \backslash R$ are proportional to their cardinalities. That is, $\Delta \cdot \frac{|R| - k_{i-1}}{n_i - k_{i-1}}$ roots are selected from $R \backslash K$, and $\Delta \cdot \frac{n_i - |R|}{n_i - k_{i-1}}$ from $V_i \backslash K$. Let the roots selected from $R \backslash K$ (resp. $V \backslash K$) be D_1 (resp. D_2), and $D = D_1 \cup D_2$ ($D \cap K = \emptyset$, $D \subseteq V_i$, and $|D| = k_i - k_{i-1} = \Delta$).

We first prove that the k_{i-1} roots from the previous round can be reused. To see it, we consider an intermediate way of root supplementation. That is, we select Δ roots from V_i / K. For any set of k_i nodes $T \subseteq V_i$, the probability that the new root set equals to T is $C_{k_i}^{k_{i-1}} / C_{n_i}^{k_{i-1}} C_{n_i - k_{i-1}}^{k_i - k_{i-1}}$. By similar derivation in Section 8.2, the probability is equal to selecting k_i fresh roots from V_i directly.

Next, we show that the probability of a node being selected as a root by Lemma 3.1 is the same as the way in Lemma 3.3, where the probability is $\Delta / (n_i - k_{i-1})$. As for our way of selecting roots from $R \backslash K$ and $V_i \backslash K$ separately, the probability of a node $u \in R \backslash K$ being selected as a root is $\frac{1}{n_i - |R|} \cdot \Delta \cdot \frac{n_i - |R|}{n_i - k_{i-1}} = \frac{\Delta}{n_i - k_{i-1}}$. For a node $u \in V_i \backslash R$, the probability is $\frac{1}{|R| - k_{i-1}} \cdot \Delta \cdot \frac{|R| - k_{i-1}}{n_i - k_{i-1}} = \frac{\Delta}{n_i - k_{i-1}}$. Thus, the probability of any node in $V_i \backslash K$ being selected as a root is the same as selecting directly from $V_i \backslash K$.

8.2 Proof of Lemma 3.3

For ease of notation, we write $\Delta = k_i - k_{i-1}$. Consider an arbitrary set K_i of k_i nodes in V_i. When generating a fresh mRR-set, the probability of K_i being selected as the roots is $1/C_{n_i}^{k_i}$, which is uniformly distributed. To establish the lemma, we only need to prove that the root set extended from the previous round is also uniformly distributed, corresponding with K_i with probability $1/C_{n_i}^{k_i}$.

Let $K_{i-1} = K_1 \cup K_2$ ($K_1 \cap K_2 = \emptyset$), where K_1 is the set of inactivated roots preserved from V_{i-1}, and K_2 is the set of roots activated and deleted in the last round. According to Alg. 4, to obtain k_i roots, we need to select $\Delta' = \Delta + |K_2|$ roots from V_i. Combining the Δ' roots and the preserved roots K_1, we obtain the new root set K_i'. The probability that K_i' corresponds with K_i is

$$\begin{aligned}
\Pr[K_i' = K_i] &= \frac{C_{k_i}^{|K_1|}}{C_{n_i}^{|K_1|} C_{n_i - |K_1|}^{\Delta'}} \\
&= \frac{k_i!}{|K_1|!(k_i - |K_1|)!} \cdot \frac{|K_1|!(n_i - |K_1|)!\Delta'!(n_i - |K_1| - \Delta')!)}{n_i!(n_i - |K_1|)!} \\
&= \frac{k_i!(n_i - k_i)}{n_i!} = 1/C_{n_i}^{k_i},
\end{aligned}$$

where the second equality is due to $\Delta' + |K_1| = k_i$.

As can be seen, after applying Alg. 4 to update the roots of an mRR-set, the distribution of the roots is the same as directly selecting nodes from the new residual graph, both uniformly distributed.

8.3 Concentration Inequalities

Let Y_i be an indicator random variable, which takes value 1 if the i-th randomly generated mRR-set is covered by the seed set S, and takes value 0 otherwise. On this basis, we present Lemma 8.1, which is proven in [40].

Lemma 8.1. *Let $\bar{Y} = \frac{1}{\theta}\sum_{i=1}^{\theta} Y_i$ denote the average of the indicator random variables. For any $\lambda \geq 0$, we have*

$$\Pr\left[\mathbb{E}[\bar{Y}]\cdot\theta < \left(\sqrt{\bar{Y}\cdot\theta + \frac{2\lambda}{9}} - \sqrt{\frac{\lambda}{2}}\right)^2 - \frac{\lambda}{18}\right] \leq e^{-\lambda},$$

$$\Pr\left[\mathbb{E}[\bar{Y}]\cdot\theta > \left(\sqrt{\bar{Y}\cdot\theta + \frac{\lambda}{2}} + \sqrt{\frac{\lambda}{2}}\right)^2\right] \leq e^{-\lambda}.$$

The derivation of Theorem 3.5 relies on Lemma 8.2 and Lemma 8.3.

Lemma 8.2. *Given a budget B and a seed set S selected by* GAAS, *if $c(S) = B$ (i.e., the budget is used up exactly), then the coverage of S approximates the seed set S^o with the largest coverage by $\Lambda(S) \geq \rho_b\Lambda(S^o)$, which holds with probability at least $1 - 1/n_i$.*

Lemma 8.3. *In each round, the seed set S returned by* GAAS *approximates the optimal seed set S^o that takes the same cost as S in the following form*

$$\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})] \geq \rho_b(1-1/e)(1-\epsilon)\mathbb{E}[\Gamma(S^o|\mathcal{S}_{i-1})].$$

To prove the approximation ratio of GAAS, we consider a dual problem of AMCSS, called truncated influence maximization (TIM). Specifically, we are given some budget B, and select b seeds in each round. The influence considered is truncated by η. In this way, we connect TIM and AMCSS. The aim is to maximize the truncated influence.

To solve TIM, we adopt the same seed selection process as GAAS in each round. Nevertheless, the stopping condition is not reaching η, but the budget is used up. For the last round of seed selection, the remaining budget B' may be not enough (i.e., $B' < c(S)$). In this case, we select S with probability $B'/c(S)$, to respect the budget constraint in expectation.

Let the policy described above be π^d. To derive a general performance guarantee for π^d, we consider a general budget l, which can be larger than B. Given the approximation ratio of $\rho_b(1-1/e)(1-\epsilon)$ in each round of seed selection by Lemma 8.3, we can derive the approximation ratio of π^d in Theorem 8.4 by applying Theorem 38 in [14], which only relies on the adaptive submodularity and adaptive monotonicity of $\Gamma(\cdot)$ which are proven in [39]. That is, we have $\mathbb{E}[\Gamma(\pi^d)] \geq (1-e^{-\alpha l/B})\mathbb{E}[\Gamma(\pi)]$, where $\alpha = \rho_b(1-1/e)(1-\epsilon)$ and π is an arbitrary policy for TIM.

Theorem 8.4. *Consider any user cost $c : V \to \mathbb{N}$ and any seed selection accuracy $0 \leq \alpha \leq 1$ in each round. For any policy π under budget B, we have*

$$\mathbb{E}[\Gamma(\pi^d)] \geq (1-e^{-\alpha l/B})\mathbb{E}[\Gamma(\pi)].$$

8.4 Proof of Lemma 8.2

By previous work [41], we know that $\Lambda(\cdot)$ is a submodular set function, since it is actually the objective function of the set cover problem. Let S^* be the node set with the largest coverage. Note that S^* is different from S^o, since the definitions of their optimality are different. And we have $\Lambda(S^*) \geq \Lambda(S^o)$. By Lemma 3 of [26], we have

$$\Lambda(S) \geq \left[1 - \prod_{j=1}^{b}\left(1 - \frac{c(s_j)}{B}\right)\right]\Lambda(S^*), \tag{1}$$

where s_j is the j-th seed selected into S. Note that the estimation error in Lemma 3 of [26] is set to 0, since the coverage of any node set in a collection of mRR-sets can be accurately counted.

By the arguments in the proof of Theorem 1 of [26], the approximation ratio in Ineq. 1 takes the minimum value when $c(s_j) = B/b, \forall j$. Thus, we have

$$\Lambda(S) \geq \left[1 - \prod_{j=1}^{b}\left(1 - \frac{1}{b}\right)\right]\Lambda(S^*) = \rho_b\Lambda(S^*) \geq \rho_b\Lambda(S^o).$$

8.5 Proof of Lemma 8.3

The proof idea is similar to the proof in [39]. We first clarify two concepts during the seed selection in GAAS, which involves multiple trials to select a seed. In each trial, GAAS *proposes* a candidate seed set, based on the coverages of nodes. If the candidate seed set is verified to be unqualified, GAAS discards this candidate set, and doubles the mRR-sets to select a new candidate set. Otherwise, GAAS *returns* the selected seed set as the final seed set.

GAAS contains two stopping conditions. First, the number of mRR-sets reaches $\theta_{\max}$; second, the candidate seed set is verified to be legitimate. We next show that, in any case, the seed set returned by GAAS is approximates the optimal seed set by at least $\rho_b(1-1/e)(1-\epsilon)$.

For the first case $\theta \geq \theta_{\max}$, following similar derivation from [39, 40], we can see that $\mathbb{E}[\hat{\Gamma}(S|\mathcal{S}_{i-1})] \geq (1-\hat{\epsilon})\rho_b\mathbb{E}[\hat{\Gamma}(S^o|\mathcal{S}_{i-1})]$ holds with probability $1-\delta/3$. Further, we have

$$\begin{aligned}&\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})] \geq \mathbb{E}[\hat{\Gamma}(S|\mathcal{S}_{i-1})]\\ \geq&(1-\hat{\epsilon})\rho_b\mathbb{E}[\hat{\Gamma}(S^o|\mathcal{S}_{i-1})] \geq (1-\hat{\epsilon})\rho_b(1-1/e)\mathbb{E}[\Gamma(S^o|\mathcal{S}_{i-1})],\end{aligned}$$

where the first and last inequalities are due to Ineq. (17).

For the second case, we consider the following two events for any seed set $S \subseteq V$,

$$E_1(S) : \mathbb{E}[\Lambda(S)] \geq \left(\sqrt{\Lambda(S) + \frac{2a_1}{9}} - \sqrt{\frac{a_1}{2}}\right)^2 - \frac{a_1}{18},$$

$$E_2(S) : \mathbb{E}[\Lambda(S)] \leq \left(\sqrt{\Lambda(S) + \frac{a_2}{2}} + \sqrt{\frac{a_2}{2}}\right)^2.$$

Then, by Lemma 8.1 and the definitions of a_1 and a_2, we have

$$\Pr[\neg E_1(S)] \leq \delta/3HC_{n_i}^b,$$
$$\Pr[\neg E_2(S)] \leq \delta/3H.$$

Note that the seed set S returned by GAAS is random, and can be any node set with size b from V. Thus, we apply the total probability

and derive

$$\begin{aligned}\Pr\left[\mathbb{E}[\Lambda(S) < \Lambda^l(S)]\right] &= \Pr[\neg E_1(S)] \\ &= \sum_{S' \subseteq V_i} \Pr[\neg E_1(S')] \cdot \Pr[S' = S] \\ &\le \frac{\delta}{3H} \sum_{S' \subseteq V_i} \Pr[S = S'] \\ &= \delta/3H.\end{aligned} \tag{2}$$

For the optimal seed set S^o, the event $E_2(S^o)$ directly implies

$$E_2'(S^o) : \mathbb{E}[\Lambda(S^o)] \le \left(\sqrt{\Lambda(S)/\rho_b + \frac{a_2}{2}} + \sqrt{\frac{a_2}{2}}\right)^2 = \Lambda^u(S^o),$$

since $\Lambda(S) \ge \rho_b \Lambda(S^o)$ by Lemma 8.2 and the r.s.h. of E_2 is increasing with $\Lambda(S^o)$. Note that S^o is unknown but certain. Thus, we do not need to apply the total probability and can write

$$\Pr[\mathbb{E}[\Lambda(S^o)] > \Lambda^u(S^o)] = \Pr[\neg E_2'(S^o)] \le \Pr[\neg E_2(S^o)] \le \delta/3H. \tag{3}$$

Combining the above analysis for S and S^o, we can see that, with probability $1 - 2\delta/3H$, the events in Ineqs. (2) and (3) do not hold, which lead to $\mathbb{E}[\Lambda(S)] \ge \Lambda^l(S)$ and $\mathbb{E}[\Lambda(S^o)] \le \Lambda^u(S^o)$. Further, when GAAS returns S in the second case, we can derive

$$\begin{aligned}&\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})] \ge \mathbb{E}[\hat{\Gamma}(S|\mathcal{S}_{i-1})] = \mathbb{E}[\Lambda(S)]n_i/\theta \ge \Lambda^l(S)n_i/\theta \\ \ge&(1-\hat{\epsilon})\rho_b\Lambda^u(S^o)n_i/\theta \ge (1-\hat{\epsilon})\rho_b\mathbb{E}[\Lambda(S^o)]n_i/\theta \\ =&(1-\hat{\epsilon})\rho_b\mathbb{E}[\hat{\Gamma}(S^o|\mathcal{S}_{i-1})] \ge (1-\hat{\epsilon})\rho_b(1-1/e)\mathbb{E}[\Gamma(S^o|\mathcal{S}_{i-1})],\end{aligned} \tag{4}$$

where the first and last inequalities are due to Ineq. (17). The second and penultimate equalities are due to the definition of $\hat{\Gamma}(\cdot)$. The third and antepenult inequalities are due to Ineqs. (2) and (3). The middle fourth inequality is due to the stopping condition of GAAS. When η is reached, at most H rounds of seed selection are carried out. The probability that Ineq. (4) holds in any round is at least $1 - (2\delta/3H) \times H = 1 - 2\delta/3$.

Although Ineq. (4) fails with very small probability, we still take it into account to derive the performance guarantee of GAAS in expectation, which is taken w.r.t. the randomness ω in seed selection. That is

$$\begin{aligned}&\mathbb{E}_\omega[\mathbb{E}[\Gamma(S|\mathcal{S}_{i-1})]] \\ \ge&(1-\hat{\epsilon})(1-\delta)\rho_b(1-1/e)\mathbb{E}[\Gamma(S^o|\mathcal{S}_{i-1})] \\ \ge&(1-\epsilon)\rho_b(1-1/e)\mathbb{E}[\Gamma(S^o|\mathcal{S}_{i-1})].\end{aligned}$$

This completes the proof.

8.6 Proof of Theorem 3.5

The proof idea is similar to [11]. We prove this theorem by showing that when we spend $(\ln\eta + 1)$ times more cost than the optimal policy, the influence threshold η can be reached with high probability.

Let π^o be the optimal policy. The expected cost of π^o is $\mathbb{E}[c(\pi^o)]$. Next, we apply Theorem 8.4 by letting $B = \mathbb{E}[c(\pi^o)]$ and $l = \frac{2\ln\eta}{(1-\epsilon)(1-1/e)\rho_b}\mathbb{E}[c(\pi^o)]$, i.e., the ratio times optimal cost. Recall the influence obtained by π^d in Theorem 8.4 and let the arbitrary policy be π^o, we see that

$$\mathbb{E}[\Gamma(\pi^d)] \ge \left(1 - e^{-\frac{\rho_b(1-1/e)(1-\epsilon)l}{\mathbb{E}[c(\pi^o)]}}\right)\mathbb{E}[\Gamma(\pi^o)].$$

Plugging the value of l, we derive that

$$\mathbb{E}[\Gamma(\pi^d)] \ge \left(1 - e^{-2\ln\eta}\right)\mathbb{E}[\Gamma(\pi^o)] = \eta - 1/\eta.$$

Note that $\mathbb{E}[\Gamma(\pi^d)]$ is an expected influence, which can be expressed as

$$\begin{aligned}\eta - 1/\eta \le \mathbb{E}[\Gamma(\pi^d)] &= \sum_{x=1} x \Pr[\Gamma(\pi^d = x)] \\ \le&\eta \Pr[\Gamma(\pi^d) = \eta] + (\eta - 1)\Pr[\Gamma(\pi^d) < \eta],\end{aligned}$$

which leads to $\Pr[\Gamma(\pi^d) = \eta] \ge 1 - 1/\eta$.

Note that the policy π^d designed for TIM selects seed sets in the same as GAAS in AMCSS, and the influence function is also truncated. Thus, the influence reached by π^d is the same as GAAS, before the last round of seed selection, where the seed set in π^d is selected probabilistically. To achieve at least the same influence as π^d, we only need to ask GAAS to select the seed set in the last round of π^d with certainty. Then, the cost spent by GAAS in the last round is also certain, resulting at most b more cost than π^d. That is, when the cost used by GAAS is $l + b$, we have $\Pr[\Gamma(\text{GAAS}) = \eta] \ge 1 - 1/\eta$.

On the other hand, with probability at most $1/\eta$, the influence of GAAS is smaller than η. In this case, at most η users are selected, resulting in a cost of η.

Summarizing the above analysis, we can derive the expected cost GAAS needed to reach η

$$\begin{aligned}\mathbb{E}[c(\text{GAAS})] &\le (l + b)(1 - 1/\eta) + \eta \cdot 1/\eta \le l + b + 1 \\ &= \frac{2\ln\eta}{(1-\epsilon)(1-1/e)\rho_b}\mathbb{E}[c(\pi^o)] + b + 1.\end{aligned}$$

This completes the proof.

9 ADDITIONAL RESULTS

Time breakdown and Effect of `ParallelUpdate`. We present a detailed time breakdown for the LT model in Figure 12. As illustrated, in GAAS (marked with ⋆), the mRR-set update phase accounts for a substantial portion of the total running time, while GAAS_{gen} spends nearly all its time on mRR-set generation. Overall, GAAS consistently achieves significantly better efficiency than GAAS_{gen}, confirming the effectiveness of our proposed mRR-set update kernel.

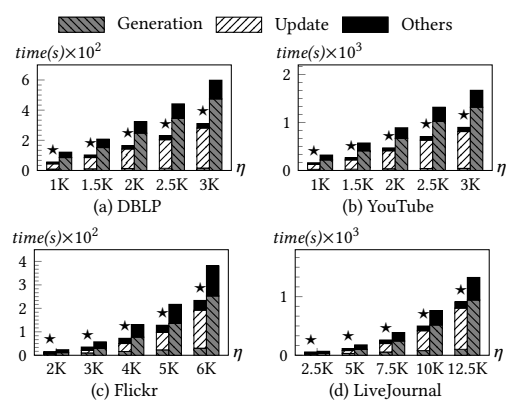


Figure 12: Runtime breakdown under the LT model; for each η, the left bar with ⋆ is GAAS and the right bar is GAAS_{gen}.

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